chapter

# 3

# MICROBIAL KINETICS

The previous chapters emphasized that microorganisms fuel their lives by performing oxidation/reduction reactions that generate the energy and reducing power needed to construct and maintain themselves. Because redox reactions are nearly always very slow unless catalyzed, microorganisms produce enzyme catalysts that increase the kinetics of their essential reactions to rates fast enough for them to exploit the chemical resources available in their environment. Engineers want to take advantage of these microbially catalyzed reactions, because the chemical resources of the microorganisms usually are the pollutants that the engineers must control. For example, the biochemical oxygen demand (BOD) is an organic electron donor for heterotrophic bacteria, $NH_4^+$-N is an inorganic electron donor for nitrifying bacteria, $NO_3^-$-N is an electron acceptor for denitrifying bacteria, and $PO_4^{3-}$ is a nutrient for all microorganisms.

In trying to employ microorganisms for pollution control, engineers must recognize two interrelated principles: First, metabolically active microorganisms catalyze the pollutant-removing reactions. The rate of pollutant removal depends on the concentration of the catalyst, or the active biomass. Second, the active biomass is grown and sustained through the utilization of its energy- and electron-generating primary substrates, which are its electron donor and electron acceptor. The rate of production of active biomass is proportional to the utilization rate of the primary substrates.

The connection between the active biomass (the catalyst) and the primary substrates is the most fundamental factor needed for understanding and exploiting microbial systems for pollution control. Because those connections must be made systematically and quantitatively for engineering design and operation, mass-balance modeling is an essential tool. That modeling is the foundation for the subject of this chapter.

## 3.1 BASIC RATE EXPRESSIONS

At a minimum, a model of a microbial process must have mass balances on the active biomass and the primary substrate that limits the growth rate of the biomass. In the vast majority of cases, the rate-limiting substrate is the electron donor. That convention is used here, and the term *substrate* now refers to the *primary electron-*

*donor substrate*. To complete the mass-balance equations, rate expressions for the growth of the biomass and utilization of the substrate must be supplied. Those two rate expressions are presented first.

The relationship most frequently used to represent bacterial growth kinetics is the so-called *Monod equation,* which was developed in the 1940s by the famous French microbiologist Jacques Monod. His original work related the specific growth rate of fast-growing bacteria to the concentration of a rate-limiting, electron-donor substrate,

$$\mu_{syn} = \left(\frac{1}{X_a}\frac{dX_a}{dt}\right)_{syn} = \hat{\mu}\frac{S}{K+S} \qquad [3.1]$$

in which

$\mu_{syn}$ = specific growth rate due to synthesis ($T^{-1}$)
$X_a$ = concentration of active biomass ($M_xL^{-3}$)
$t$ = time (T)
$S$ = concentration of the rate-limiting substrate ($M_sL^{-3}$)
$\hat{\mu}$ = maximum specific growth rate ($T^{-1}$)
$K$ = concentration giving one-half the maximum rate ($M_sL^{-3}$)

This equation is a convenient mathematical representation for a smooth transition from a first-order relation (in $S$) at low concentration to a zero-order relation (in $S$) at high concentration. The Monod equation is sometimes called a saturation function, because the growth rate saturates at $\hat{\mu}$ for large $S$. Figure 3.1 shows how $\mu$ varies with $S$ and that $\mu = \hat{\mu}/2$ when $K = S$. Although Equation 3.1 is largely empirical, it has widespread applicability for microbial systems.

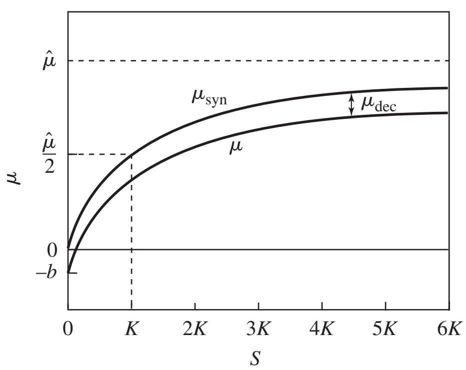


**Figure 3.1** Schematic of how the synthesis and net specific growth rates depend on the substrate concentration. At $S = 20K$, $\mu_{syn} = 0.95\hat{\mu}$.

People who studied more slowly growing bacteria (environmental engineers are among the key examples of these people) discovered that active biomass has an energy demand for maintenance, which includes cell functions such as motility, repair and resynthesis, osmotic regulation, transport, and heat loss. Environmental engineers usually represent that flow of energy and electrons required to meet maintenance needs as *endogenous decay*. In other words, the cells oxidize themselves to meet maintenance-energy needs.

The rate of endogenous decay is

$$\mu_{\text{dec}} = \left(\frac{1}{X_a}\frac{dX_a}{dt}\right)_{\text{decay}} = -b \tag{3.2}$$

in which

$b$ = endogenous-decay coefficient ($T^{-1}$)
$\mu_{\text{dec}}$ = specific growth rate due to decay ($T^{-1}$).

Equation 3.2 says that the loss of active biomass is a first-order function. However, not all of the active biomass lost by decay is actually oxidized to generate energy for maintenance needs. Although most of the decayed biomass is oxidized, a small fraction accumulates as inert biomass. The rate of oxidation (or true respiration for energy generation) is

$$\left(\frac{1}{X_a}\frac{dX_a}{dt}\right)_{\text{resp}} = -f_d b \tag{3.3}$$

in which $f_d$ = fraction of the active biomass that is biodegradable. The rate at which active biomass is converted to inert biomass is the difference between the overall decay rate and the oxidation decay rate,

$$-\frac{1}{X_a}\frac{dX_i}{dt} = \left(\frac{1}{X_a}\frac{dX_a}{dt}\right)_{\text{inert}} = -(1 - f_d)b \tag{3.4}$$

in which $X_i$ = inert biomass concentration ($M_xL^{-3}$).

Overall, the net specific growth rate of active biomass ($\mu$) is the sum of new growth (Equation 3.1) and decay (Equation 3.2):

$$\mu = \frac{1}{X_a}\frac{dX_a}{dt} = \mu_{\text{syn}} + \mu_{\text{dec}} = \hat{\mu}\frac{S}{K+S} - b \tag{3.5}$$

Figure 3.1 also shows how $\mu$ varies with $S$ and that $\mu$ can be negative for low enough $S$.

Because of our ultimate interest in substrate removal (e.g., BOD) and because biomass growth is "fueled" by substrate utilization, environmental engineers often prefer to regard the rate of substrate utilization as the basic rate, while cell growth is derived from substrate utilization. Then, the Monod equation takes the form

$$r_{\text{ut}} = -\frac{\hat{q}S}{K+S}X_a \tag{3.6}$$

in which

$r_{ut}$ = rate of substrate utilization ($M_sL^{-3}T^{-1}$)
$\hat{q}$ = maximum specific rate of substrate utilization ($M_sM_x^{-1}T^{-1}$)

Substrate utilization and biomass growth are connected by

$$\hat{\mu} = \hat{q}Y \tag{3.7}$$

in which $Y$ = true yield for cell synthesis ($M_xM_s^{-1}$). Yes, the true yield is exactly the same $Y$ as you studied in Chapter 2 on *Stoichiometry*. It represents the fraction of electron-donor electrons converted to biomass electrons during synthesis of new biomass. The net rate of cell growth becomes

$$r_{\text{net}} = Y\frac{\hat{q}S}{K+S}X_a - bX_a \tag{3.8}$$

in which $r_{\text{net}}$ = the net rate of active-biomass growth ($M_xL^{-3}T^{-1}$). Of course,

$$\mu = r_{\text{net}}/X_a = Y\frac{\hat{q}S}{K+S} - b \tag{3.9}$$

Some prefer to think of cell maintenance as being a shunting of substrate-derived electrons and energy directly for maintenance. This is expressed by

$$\mu = Y\left(\frac{\hat{q}S}{K+S} - m\right) \tag{3.10}$$

in which $m$ = maintenance-utilization rate of substrate ($M_sM_x^{-1}T^{-1}$), which also was introduced in Chapter 2. When systems go to steady state, there is no difference in the two approaches to maintenance, and $b = Ym$. We use the endogenous-decay approach exclusively.

## 3.2 PARAMETER VALUES

The parameters describing biomass growth and substrate utilization cannot be taken as "random variables." They have specific units and ranges of values. In several cases, the values are constrained by the cell's stoichiometry and energetics, described in Chapter 2.

The true yield, $Y$, is proportional to $f_s^o$, presented under *Stoichiometry*. The thermodynamic method presented at the end of Chapter 2 is an excellent means to obtain a good first estimate of $Y$ if experimental values are not available. Table 3.1 lists $f_s^o$ and $Y$ values for the most commonly encountered bacteria in environmental biotechnology. Table 3.1 illustrates that $f_s^o$ ranges from its highest values (0.6 to 0.7 $e^-$ eq cells/$e^-$ eq donor) for aerobic heterotrophs to its lowest values (0.05 to 0.10 $e^-$ eq cells/$e^-$ eq donor) for autotrophs and acetate-oxidizing anaerobes. These values reflect the balancing of energy costs for synthesis with the energy gains from

**Table 3.1** Typical $f_s^0$, $Y$, $\hat{q}$, and $\hat{\mu}$ values for key bacterial types in environmental biotechnology

| Organism Type | Electron Donor | Electron Acceptors | C-Source | $f_s^0$ | $Y$ | $\hat{q}$ | $\hat{\mu}$ |
|---|---|---|---|---|---|---|---|
| Aerobic, Heterotrophs | Carbohydrate BOD | $O_2$ | BOD | 0.7 | 0.49 gVSS/g$BOD_L$ | 27 g$BOD_L$/gVSS-d | 13.2 |
| | Other BOD | $O_2$ | BOD | 0.6 | 0.42 gVSS/g$BOD_L$ | 20 g$BOD_L$/gVSS-d | 8.4 |
| Denitrifiers | BOD | $NO_3^-$ | BOD | 0.5 | 0.25 gVSS/g$BOD_L$ | 16 g$BOD_L$/gVSS-d | 4 |
| | $H_2$ | $NO_3^-$ | $CO_2$ | 0.2 | 0.81 gVSS/g$H_2$ | 1.25 g$H_2$/gVSS-d | 1 |
| | S(s) | $NO_3^-$ | $CO_2$ | 0.2 | 0.15 gVSS/gS | 6.7 gS/gVSS-d | 1 |
| Nitrifying Autotrophs | $NH_4^+$ | $O_2$ | $CO_2$ | 0.14 | 0.34 gVSS/g$NH_4^+$-N | 2.7 g$NH_4^+$-N/gVSS-d | 0.92 |
| | $NO_2^-$ | $O_2$ | $CO_2$ | 0.10 | 0.08 gVSS/g$NO_2^-$-N | 7.8 g$NO_2^-$-N/gVSS-d | 0.62 |
| Methanogens | acetate BOD | acetate | acetate | 0.05 | 0.035 gVSS/g$BOD_L$ | 8.4 g$BOD_L$/gVSS-d | 0.3 |
| | $H_2$ | $CO_2$ | $CO_2$ | 0.08 | 0.45 gVSS/g$H_2$ | 1.1 g$H_2$/g VSS-d | 0.5 |
| Sulfide Oxidizing Autotrophs | $H_2S$ | $O_2$ | $CO_2$ | 0.2 | 0.28 gVSS/g$H_2$S-S | 5 gS/gVSS-d | 1.4 |
| Sulfate Reducers | $H_2$ | $SO_4^{2-}$ | $CO_2$ | 0.05 | 0.28 gVSS/g$H_2$ | 1.05 g$H_2$/gVSS-d | 0.29 |
| | acetate BOD | $SO_4^{2-}$ | acetate | 0.08 | 0.057 gVSS/g$BOD_L$ | 8.7 g$BOD_L$/gVSS-d | 0.5 |
| Fermenters | sugar BOD | sugars | sugars | 0.18 | 0.13 gVSS/g$BOD_L$ | 9.8 g$BOD_L$/gVSS-d | 1.2 |

$Y$ is computed assuming a cellular $VSS_a$ composition of $C_5H_7O_2N$, and $NH_4^+$ is the N source, except when $NO_3^-$ is the electron acceptor; then $NO_3^-$ is the N source. The typical units on $Y$ are presented.

$\hat{q}$ is computed using $\hat{q} = 1e^-$ eq/g$VSS_a$-d.

$\hat{\mu}$ has units of $d^{-1}$.

the donor-to-acceptor energy reaction. Chapter 2 showed that $Y$ values are computed directly from $f_s^o$ as a unit conversion. For example,

Aerobic heterotrophs:

$$Y = 0.6\frac{\text{e}^- \text{ eq cells}}{\text{e}^- \text{ eq donor}} \cdot \frac{113 \text{ gVSS}}{20 \text{ e}^- \text{ eq cells}} \cdot \frac{1 \text{ e}^- \text{ eq donor}}{8 \text{ gBOD}_\text{L}}$$
$$= 0.42 \text{ gVSS/gBOD}_\text{L}$$

Denitrifying heterotrophs:

$$Y = 0.5\frac{\text{e}^- \text{ eq cells}}{\text{e}^- \text{ eq donor}} \cdot \frac{113 \text{ gVSS}}{28 \text{ e}^- \text{ eq cells}} \cdot \frac{1 \text{ e}^- \text{ eq donor}}{8 \text{ gBOD}_\text{L}}$$
$$= 0.25 \text{ gVSS/gBOD}_\text{L}$$

$H_2$-Oxidizing Sulfate Reducers:

$$Y = 0.05\frac{\text{e}^- \text{ eq cells}}{\text{e}^- \text{ eq donor}} \cdot \frac{113 \text{ gVSS}}{20 \text{ e}^- \text{ eq cells}} \cdot \frac{2 \text{ e}^- \text{ eq donor}}{2 \text{ gH}_2}$$
$$= 0.28 \text{ gVSS/gH}_2$$

The wide range of $Y$ values reflects a combination of changes in $f_s^o$ values and different units for the donor.

The true yield also can be estimated experimentally from batch growth. A small inoculum is grown to exponential phase and harvested. The true yield is estimated from $Y = -\Delta X/\Delta S$, where $\Delta X$ and $\Delta S$ are the measured changes in biomass and substrate concentration from inoculation until the time of harvesting. The batch technique is adequate for rapidly growing cells, but can create errors when the cells grow slowly so that biomass decay cannot be neglected.

For the cells' usual primary substrates, the maximum specific rate of substrate utilization, $\hat{q}$, is controlled largely by electron flow to the electron acceptor. For 20 °C, the maximum flow to the energy reaction is about 1 e$^-$ eq/gVSS-d. If this flow is termed $\hat{q}_e$, $\hat{q}$ can be computed from

$$\hat{q} = \hat{q}_e/f_e^0 \qquad \textbf{[3.11]}$$

Table 3.1 also lists typical $\hat{q}$ values when $\hat{q}_e = 1$ e$^-$ eq/gVSS-d. The range of $\hat{q}$ values again reflects variations in $f_e^o$ and different units for the donor. Table 3.1 also lists $\hat{\mu} = Y\hat{q}$ values. The table makes it clear that the fast growing cells are those that have a large $f_s^o$, which directly gives a large $Y$ and indirectly gives a large $\hat{q}$. Thus, $\hat{\mu}$ is controlled mainly by the microorganism's stoichiometry and energetics.

Temperature affects $\hat{q}$. For temperatures up to the microorganism's optimal temperature, the substrate-utilization rate roughly doubles for each 10 °C increase in temperature. This phenomenon can be approximated by

$$\hat{q}_T = \hat{q}_{20}(1.07)^{T-20} \qquad \textbf{[3.12]}$$

where $T$ is in °C and $\hat{q}_{20}$ is the $\hat{q}$ value for 20 °C. If $\hat{q}_{20}$ is not known, the relationship can be generalized to

$$\hat{q}_T = \hat{q}_{T^R}(1.07)^{(T-T^R)} \qquad [3.13]$$

in which $T^R$ is any reference temperature (°C) for which $\hat{q}_{T^R}$ is known.

The endogenous decay rate ($b$) depends on species type and temperature. The $b$ values tend to correlate positively with $\hat{\mu}$ values. For example, aerobic heterotrophs have $b$ values of 0.1 to 0.3/d at 20 °C, while the slower-growing species have $b <$ 0.05/d. The temperature effect on $b$ can be expressed by a $(1.07)^{T-T^R}$ relationship parallel to Equation 3.13. Endogenous decay coefficients normally encompass several loss phenomena, including lysis, predation, excretion of soluble materials, and death.

McCarty (1975) found that the biodegradable fraction ($f_d$) is quite reproducible and has a value near 0.8 for a wide range of microorganisms.

The Monod half-maximum-rate concentration ($K$) is the most variable and least predictable parameter. Its value can be affected by the substrate's affinity for transport or metabolic enzymes. In addition, mass-transport resistances, usually ignored for suspended growth, often are "lumped" into the Monod kinetics by an increase in $K$. When mass transport is not included in $K$ and simple electron-donor substrates are considered, $K$ values tend to be low, less than 1 mg/l and often as low as the $\mu$g/l range. For more difficult to degrade compounds and when mass-transport resistance is de facto included, $K$ values range from a few mg/l to 100s of mg/l for electron donors. Terminal electron acceptors for respiration often have very low $K$ values, well below 1 mg/l. On the other hand, direct use of $O_2$ as a cosubstrate in oxygenation reactions may have a higher $K$ value, around 1 mg/l. The information provided here for $K$ should be taken as general guidance only. More exact values require analyzing experimental results.

## 3.3 BASIC MASS BALANCES

Writing mass balances requires specifying a control volume. All the mass balances, their solutions, and important trends can be obtained by considering one of the simplest systems: a steady-state chemostat. Figure 3.2 illustrates the key features of a chemostat: a completely mixed reactor having uniform and steady concentrations of active cells ($X_a$), substrate ($S$), inert biomass ($X_i$), and any other constituents we wish to consider. Substrate ($S$) includes soluble compounds or suspended materials that are hydrolyzed to soluble compounds within the biological reactor. The chemostat's volume is $V$, and it receives a constant feed flow rate $Q$ having substrate concentration $S^0$. For now, we preclude inputs of active or inert biomass. The effluent has flow rate $Q$ and concentrations $X_a$, $X_i$, and $S$.

We must provide mass balances on the active biomass and the rate-limiting substrate (assumed to be the electron donor), because they are the active catalysts and the material responsible for the accumulation of the catalysts, respectively. The

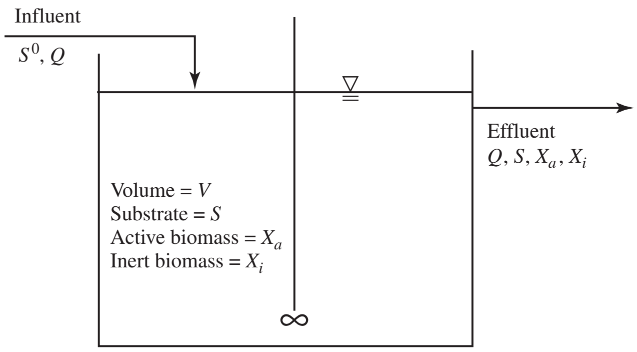


**Figure 3.2** Schematic of a chemostat.

steady-state mass balances are:

$$\text{Active Biomass:} \quad 0 = \mu X_a V - Q X_a \tag{3.14}$$

$$\text{Substrate:} \quad 0 = r_{ut} V + Q(S^0 - S) \tag{3.15}$$

Equations 3.6 and 3.9 provide the functions for $\mu$ and $r_{ut}$, respectively. Substituting them gives

$$0 = Y \frac{\hat{q} S}{K + S} X_a V - b X_a V - Q X_a \tag{3.16}$$

for biomass and

$$0 = -\frac{\hat{q} S}{K + S} X_a V + Q(S^0 - S) \tag{3.17}$$

for substrate.

Equations 3.16 and 3.17 can be solved to yield the steady-state values of $S$ and $X_a$. The strategy is to solve Equation 3.16, the active-biomass balance, first for $S$.

$$S = K \frac{1 + b\left(\frac{V}{Q}\right)}{Y\hat{q}\left(\frac{V}{Q}\right) - \left(1 + b\left(\frac{V}{Q}\right)\right)} \tag{3.18}$$

Once we know $S$, we solve the substrate balance, Equation 3.17. To do this, we first rearrange Equation 3.16 to obtain $[\hat{q} S X_a V/(K + S)]$, which is substituted into Equation 3.17 to yield

$$X_a = Y(S^0 - S) \frac{1}{1 + b\left(\frac{V}{Q}\right)} \tag{3.19}$$

This two-step strategy is a generally useful one and can be used for more complicated systems or when the growth and substrate-utilization relationships differ from those in Equations 3.6 and 3.9.

The ratio $V/Q$ is not the most convenient form. It can be substituted immediately by either of two alternate forms:

$$\text{hydraulic detention time } (T) = \theta = V/Q \qquad \textbf{[3.20]}$$

or

$$\text{dilution rate } (T^{-1}) = D = Q/V \qquad \textbf{[3.21]}$$

Engineers generally are more comfortable with using $\theta$ to simplify Equations 3.18 and 3.19, while microbiologists usually use $D$.

A much more general and powerful substitution is the *solids retention time (SRT),* which also is commonly called the *mean cell residence time (MCRT)* or the *sludge age*. These three terms normally refer to the same quantity, which is denoted $\theta_x$, has units of time, and always is defined as

$$\theta_x = \frac{\text{active biomass in the system}}{\text{production rate of active biomass}} = \mu^{-1} \qquad \textbf{[3.22]}$$

Equation 3.22 illustrates the two critical features of $\theta_x$. First, it is the reciprocal of the net specific growth rate. Thus, $\theta_x$ is a fundamental descriptor of the physiological status of the system, because it gives us direct information about the specific growth rate of the microorganisms. Second, the word definition in Equation 3.22 must be converted to quantitative parameters to apply $\theta_x$ to any system.

For our chemostat, any biomass produced in the system must exit in the effluent: thus, the denominator is $QX_a$. The total active biomass (the numerator) is $VX_a$. Thus,

$$\theta_x = \frac{VX_a}{QX_a} = \theta = \frac{1}{D} \qquad \textbf{[3.23]}$$

which emphasizes that $\theta_x = \theta$ in our chemostat, as long as the chemostat is at steady state.

Rewriting Equations 3.18 and 3.19 in the (most useful) $\theta_x$ form gives

$$S = K\frac{1 + b\theta_x}{Y\hat{q}\theta_x - (1 + b\theta_x)} \qquad \textbf{[3.24]}$$

$$X_a = Y\left(\frac{S^0 - S}{1 + b\theta_x}\right) \qquad \textbf{[3.25]}$$

Figure 3.3 illustrates how $S$ and $X_a$ are controlled by $\theta_x$. Although the chemostat is a simple system, several key trends shown in Figure 3.3 occur for all suspended-growth processes.

1. When $\theta_x$ is very small, $S = S^0$, and $X_a = 0$. This situation, termed *washout,* has no substrate removal and, therefore, no accumulation of active biomass. The $\theta_x$ value at which washout begins is called $\theta_x^{\min}$, which forms the boundary between having steady-state biomass and washout. $\theta_x^{\min}$ is computed by letting $S = S^0$ in Equation 3.24 and solving for $\theta_x$:

$$\theta_x^{\min} = \frac{K + S^0}{S^0(Y\hat{q} - b) - bK} \qquad \textbf{[3.26]}$$

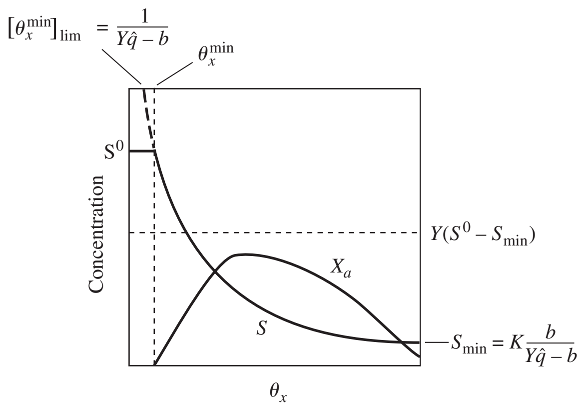


**Figure 3.3** Sketch of how $S$ and $X_a$ vary with $\theta_x$ and what are the limiting values for $\theta_x$, $S$, and $X_a$.

$\theta_x^{min}$ increases with increasing $S^0$, but asymptotically reaches a limiting value

$$\left[\theta_x^{min}\right]_{lim} = \frac{1}{Y\hat{q} - b} \quad \textbf{[3.27]}$$

which defines an absolute minimum $\theta_x$ (or maximum $\mu$) boundary for having steady-state biomass. $[\theta_x^{min}]_{lim}$ is a fundamental delimiter of a biological process: washout at low $\theta_x$.

2. For all $\theta_x > \theta_x^{min}$, $S$ declines monotonically with increasing $\theta_x$. Equation 3.24 is used to compute $S$.
3. For very large $\theta_x$, $S$ approaches another key limiting value: $S_{min}$, the minimum substrate concentration capable of supporting steady-state biomass. $S_{min}$ can be computed by letting $\theta_x$ approach infinity in Equation 3.24:

$$S_{min} = K\frac{b}{Y\hat{q} - b} \quad \textbf{[3.28]}$$

If $S < S_{min}$, the cells net specific growth rate is negative (recall Equation 3.9), and biomass will not accumulate or will gradually disappear. Therefore, steady-state biomass can be sustained only when $S > S_{min}$. $S_{min}$ is a fundamental delimiter of biological process performance for large $\theta_x$.

4. When $\theta_x > \theta_x^{min}$, $X_a$ rises initially, because $S^0 - S$ increases as $\theta_x$ becomes larger. However, $X_a$ reaches a maximum value in a chemostat and then declines as decay becomes dominant for large $\theta_x$. If $\theta_x$ were to extend to infinity, $X_a$ would approach zero.

In summary, the simple graph in Figure 3.3 tells us that we can reduce $S$ from $S^0$ to $S_{min}$ as we increase $\theta_x$ from $\theta_x^{min}$ to infinity. The exact value of $\theta_x$ we pick depends on a balancing of substrate removal, biomass production (equals $QX_a$), and other factors we will discuss later. In practice, engineers often specify a microbiological safety factor, which is defined as $\theta_x/\theta_x^{min}$. Safety factors, which typically range from around five to hundreds, are discussed in Chapter 5.

## 3.4 MASS BALANCES ON INERT BIOMASS AND VOLATILE SOLIDS

Because some fraction of newly synthesized biomass is refractory to self-oxidation, endogenous respiration leads to the accumulation of inactive biomass. In addition, real influents often contain refractory volatile suspended solids that we cannot differentiate easily from inactive biomass. So, we must expand our chemostat analysis to account for inactive biomass (and other nonbiodegradable volatile suspended solids).

A steady-state mass balance on inert biomass is

$$0 = (1 - f_d)bX_aV + Q(X_i^0 - X_i) \quad \textbf{[3.29]}$$

in which

$X_i$ = concentration of inert biomass in the chemostat ($M_xL^{-3}$)

$X_i^0$ = input concentration of inert biomass (or indistinguishable refractory volatile suspended solids) ($M_xL^{-3}$).

Thus, we are relaxing the original requirement that only substrate enters in the influent. Also, the first term in Equation 3.29, the formation rate of inert biomass from active-biomass decay, comes from Equation 3.4.

Solution of Equation 3.29 is

$$X_i = X_i^0 + X_a(1 - f_d)b\theta \quad \textbf{[3.30]}$$

Equation 3.30 emphasizes that $X_i$ is comprised of influent inerts ($X_i^0$) and inerts formed from decay of $X_a$ (i.e., $X_a(1 - f_d)b\theta$). Figure 3.4 shows that $X_i$ increases monotonically from $X_i^0$ to a maximum of $X_i^0 + Y(S^0 - S_{min})(1 - f_d)$. Thus, operation at a large $\theta$ results in greater accumulation of inert biomass.

The sum of $X_i$ and $X_a$ is called $X_v$, the volatile suspended solids concentration, or VSS. Letting $\theta_x = \theta$ for the chemostat, $X_v$ can be computed directly as

$$X_v = X_i^o + X_a(1 + (1 - f_d)b\theta_x) = X_i^0 + Y(S^0 - S)\frac{1 + (1 - f_d)b\theta_x}{1 + b\theta_x} \quad \textbf{[3.31]}$$

Figure 3.4 demonstrates that $X_v$ generally follows the trend of $X_a$, but it does not equal zero; when $X_a$ goes to zero, $X_v$ equals $X_i$.

The second term on the right-hand side of Equation 3.31 represents the net accumulation of biomass from synthesis and decay. It is constituted by the change in substrate concentration ($S^0 - S$) multiplied by the net yield ($Y_n$):

$$Y_n = Y\frac{1 + (1 - f_d)b\theta_x}{1 + b\theta_x} \quad \textbf{[3.32]}$$

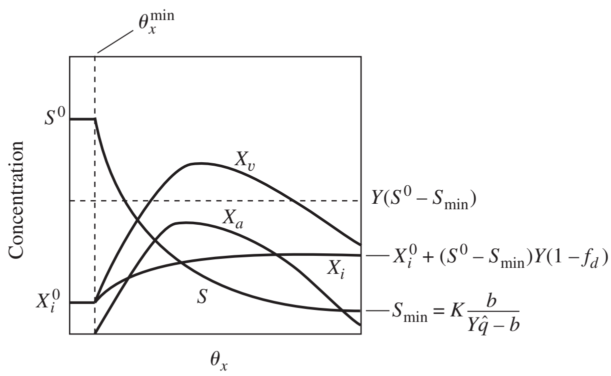


**Figure 3.4** General variations when $X_i$ and $X_v$ are added to the chemostat analysis.

Equation 3.32 is parallel to the relationship between $f_s$ and $f_s^0$:

$$f_s = f_s^0 \frac{1 + (1 - f_d)b\theta_x}{1 + b\theta_x} \quad \textbf{[3.33]}$$

Sometimes the net yield also is given the name observed yield and symbolized as $Y_{obs}$.

## 3.5 SOLUBLE MICROBIAL PRODUCTS

Besides consuming substrates and making new biomass, bacteria also generate *soluble microbial products (SMP)*. SMP ($M_pL^{-3}$) appear to be cellular components that are released during cell lysis, diffuse through the cell membrane, are lost during synthesis, or are excreted for some purpose. They have moderate formula weights (100s to 1,000s) and are biodegradable. SMP generally do not include intermediates of catabolic pathways, which behave differently than SMP and are system specific when they occur. SMP are important because they are present in all cases and form the majority of the effluent COD and BOD in many cases. SMP also can complex metals, foul membranes, and cause color or foaming.

SMP can be subdivided into two subcategories. *UAP* (short for *substrate-utilization-associated products*) are produced directly during substrate metabolism. Their formation kinetics can be described by

$$r_{UAP} = -k_1 r_{ut} \quad \textbf{[3.34]}$$

in which

$r_{UAP}$ = rate of UAP-formation $M_pL^{-3}T^{-1}$
$k_1$ = UAP-formation coefficient, $M_pM_s^{-1}$

The second category is *BAP* (for *biomass-associated products*). They are formed directly from biomass, presumably as part of maintenance and decay, but perhaps for other reasons, as well. The BAP formation rate expression is

$$r_{\mathrm{BAP}} = k_2 X_a \qquad [3.35]$$

in which

$r_{\mathrm{BAP}}$ = rate of BAP formation, $\mathrm{M}_p\mathrm{L}^{-3}\mathrm{T}^{-1}$

$k_2$ = BAP-formation rate coefficient, $\mathrm{M}_p\mathrm{M}_x^{-1}\mathrm{T}^{-1}$

From several lines of research (summarized in Rittmann, Bae, Namkung, and Lu (1987) and Noguera, Araki, and Rittmann (1994)), we know that SMP originally formed is relatively biodegradable. However, being a highly heterogeneous mixture, the SMP has a range of biodegradation kinetics. The details of SMP-biodegradation kinetics are not fully resolved. However, the most recent work suggests that UAP and BAP have sufficiently distinct degradation kinetics that they can be described with separate Monod-degradation expressions:

$$r_{\mathrm{deg-UAP}} = \frac{-\hat{q}_{\mathrm{UAP}}\mathrm{UAP}}{K_{\mathrm{UAP}} + \mathrm{UAP}} X_a \qquad [3.36a]$$

$$r_{\mathrm{deg-BAP}} = \frac{-\hat{q}_{\mathrm{BAP}}\mathrm{BAP}}{K_{\mathrm{BAP}} + \mathrm{BAP}} X_a \qquad [3.36b]$$

in which $\hat{q}_{\mathrm{UAP}}$ and $\hat{q}_{\mathrm{BAP}}$ are maximum specific rates of UAP and BAP degradation ($\mathrm{M}_p\mathrm{M}_x^{-1}\mathrm{T}^{-1}$), respectively; $K_{\mathrm{UAP}}$ and $K_{\mathrm{BAP}}$ are half-maximum rate concentrations for UAP and BAP ($\mathrm{M}_p\mathrm{L}^{-3}$), respectively; and UAP and BAP stand for their concentrations ($\mathrm{M}_p\mathrm{L}^{-3}$). Note that only active biomass ($X_a$) degrades SMP.

The steady-state mass balances on UAP and BAP are

$$0 = -k_1 r_{\mathrm{ut}} V - \frac{\hat{q}_{\mathrm{UAP}}\mathrm{UAP}}{K_{\mathrm{UAP}} + \mathrm{UAP}} X_a V - Q\mathrm{UAP} \qquad [3.37a]$$

$$0 = k_2 X_a V - \frac{\hat{q}_{\mathrm{BAP}}\mathrm{BAP}}{K_{\mathrm{BAP}} + \mathrm{BAP}} X_a V - Q\mathrm{BAP} \qquad [3.37b]$$

These are solved for UAP and BAP as

$$\mathrm{UAP} = -\frac{(\hat{q}_{\mathrm{UAP}} X_a \theta + K_{\mathrm{UAP}} + k_1 r_{\mathrm{ut}} \theta)}{2} + \frac{\sqrt{(\hat{q}_{\mathrm{UAP}} X_a \theta + K_{\mathrm{UAP}} + k_1 r_{\mathrm{ut}} \theta)^2 - 4 K_{\mathrm{UAP}} k_1 r_{\mathrm{ut}} \theta}}{2} \qquad [3.38]$$

$$\mathrm{BAP} = \frac{-(K_{\mathrm{BAP}} + (\hat{q}_{\mathrm{BAP}} - k_2) X_a \theta)}{2} + \frac{\sqrt{(K_{\mathrm{BAP}} + (\hat{q}_{\mathrm{BAP}} - k_2) X_a \theta)^2 + 4 K_{\mathrm{BAP}} k_2 X_a \theta}}{2} \qquad [3.39]$$

In Equations 3.38 and 3.39, the hydraulic detention time ($\theta$) appears, and $r_{\mathrm{ut}}$ takes a negative value and can be computed from

$$r_{\mathrm{ut}} = -(S^0 - S)/\theta = -\frac{\hat{q} S}{K + S} X_a \qquad [3.40]$$

Information to define the SMP kinetic parameters is sparse. Noguera (1991) analyzed aerobic data and obtained the following best-fit values:

$$\begin{aligned}
k_1 &= 0.12 \text{ g COD}_p/\text{g COD}_s \\
k_2 &= 0.09 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
\hat{q}_{\text{UAP}} &= 1.8 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
K_{\text{UAP}} &= 100 \text{ mg COD}_p/\text{l} \\
\hat{q}_{\text{UAP}}/K_{\text{UAP}} &= 18 \text{ l/g VSS}_a\text{-d} \\
\hat{q}_{\text{BAP}} &= 0.1 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
K_{\text{BAP}} &= 85 \text{ mg COD}_p/1 \\
\hat{q}_{\text{BAP}}/K_{\text{BAP}} &= 1.2 \text{ l/g VSS}_a\text{-d}
\end{aligned}$$

While these values can be considered only provisional, they do point out several key features of SMP. First, a significant fraction of substrate COD is shunted to UAP formation; $k_1 = 0.12$ g $\text{COD}_p$/mg $\text{COD}_s$ means that 12 percent of the substrate "utilized" is neither sent to the electron acceptor nor converted to biomass, but is released as UAP. Second, the formation of BAP constitutes a significant fraction of biomass loss; $k_2 = 0.09$ g $\text{COD}_p$/g $\text{VSS}_a$-d converts to a first-order "decay" rate of approximately 0.06/d. Third, UAP is biodegraded considerably faster than is BAP; thus, we should expect to see preferential buildup of BAP in most situations.

Noguera et al. (1994) estimated SMP parameters for a methanogenic system:

$$\begin{aligned}
k_1 &= 0.21 \text{ g COD}_p/\text{g COD}_s \\
k_2 &= 0.035 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
\hat{q}_{\text{UAP}}/K_{\text{UAP}} &= 2.4 \text{ l/g VSS}_a\text{-d} \\
\hat{q}_{\text{BAP}}/K_{\text{BAP}} &= 0.31 \text{ l/g VSS}_a\text{-d}
\end{aligned}$$

Most notable is the relatively slower degradation kinetics, compared to the aerobic system.

Since $k_2$ can be a significant fraction of $b$, we shall assume that $b$ implicitly includes BAP generation, as well as biomass oxidation and conversion to inert biomass. By differences, the portion of endogenous decay that results in direct biomass oxidation is $b - f_d b - k_2 = (1 - f_d)b - k_2$.

**Example 3.1** **ANALYSIS OF A CHEMOSTAT'S PERFORMANCE** A chemostat having volume $V = 2{,}000 \text{ m}^3$ receives a constant flow of $Q = 1{,}000 \text{ m}^3$/d of wastewater containing only biodegradable organic material having $\text{BOD}_L$ of $S^0 = 500$ mg $\text{BOD}_L$/l. Also included is inert VSS with $X_i^0 = 50$ mg VSS/l. From past research, we know that the electron donor is rate limiting, and we have the following kinetic and stoichiometric parameters for aerobic biodegradation:

$$\begin{aligned}
\hat{q} &= 20 \text{ g BOD}_L/\text{g VSS}_a\text{-d} & k_2 &= 0.09 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
Y &= 0.42 \text{ g VSS}_a/\text{g BOD}_L & \hat{q}_{\text{UAP}} &= 1.8 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
K &= 20 \text{ mg BOD}_L/\text{l} & K_{\text{UAP}} &= 100 \text{ mg COD}_p/\text{l} \\
b &= 0.15/\text{d} & \hat{q}_{\text{BAP}} &= 0.1 \text{ g COD}_p/\text{g VSS}_a\text{-d} \\
f_d &= 0.8 & K_{\text{BAP}} &= 85 \text{ mg COD}_p/\text{l} \\
k_1 &= 0.12 \text{ g COD}_p/\text{g BOD}_L
\end{aligned}$$

The goal is to analyze the performance of the chemostat with regard to its effluent quality. First, we compute the delimiting values of $S_{\min}$, $[\theta_x^{\min}]_{\lim}$, and $\theta_x^{\min}$:

$$S_{\min} = K\frac{b}{Y\hat{q} - b} = \frac{20 \text{ mg BOD}_L}{1}\frac{0.15/\text{d}}{0.42\dfrac{\text{g VSS}_a}{\text{g BOD}_L}\cdot 20\dfrac{\text{g BOD}_L}{\text{g VSS}_a\text{-d}} - 0.15/\text{d}}$$

$$= 0.36 \text{ mg BOD}_L/\text{l}$$

$$[\theta_x^{\min}]_{\lim} = \frac{1}{Y\hat{q} - b} = \frac{1}{0.42\dfrac{\text{g VSS}_a}{\text{g BOD}_L}\cdot 20\dfrac{\text{g BOD}_L}{\text{g VSS}_a\text{-d}} - 0.15/\text{d}}$$

$$= 0.12 \text{ d}$$

$$\theta_x^{\min} = \frac{K + S^0}{S^0(Y\hat{q} - b) - bK}$$

$$= \frac{(20 + 500) \text{ mg BOD}_L/\text{l}}{500\dfrac{\text{mg BOD}_L}{1}(0.42 \cdot 20/\text{d} - 0.15/\text{d}) - 20 \text{ mg BOD}_L/\text{l} \cdot 0.15/\text{d}}$$

$$= \frac{520 \text{ mg BOD}_L/\text{l}}{(4{,}125 - 3) \text{ mg BOD}_L/\text{l-d}}$$

$$= 0.126 \text{ d}$$

We see that the substrate concentration could be driven well below $S^0$, as long as the SRT is significantly greater than 0.126 d. Note also that the product $Y\hat{q} = \hat{\mu} = 8.4 \text{ d}^{-1}$ recurs frequently.

Second, we compute $\theta_x$ and the safety factor (SF)

$$\theta_x = \theta = V/Q = \frac{2{,}000 \text{ m}^3}{1{,}000 \text{ m}^3/\text{d}} = 2 \text{ d}$$

$$\text{SF} = \theta_x/\theta_x^{\min} = \frac{2 \text{ d}}{0.126 \text{ d}} = 16$$

We should have $\theta_x$ and SF large enough to take $S$ much below $S^0$ and to give stable biomass accumulation.

Third, we compute the effluent substrate concentration.

$$S = K\frac{1 + b\theta_x}{Y\hat{q}\theta_x - (1 + b\theta_x)}$$

$$= 20\frac{\text{mg BOD}_L}{1}\frac{1 + 0.15/\text{d} \cdot 2 \text{ d}}{8.4/\text{d} \cdot 2 \text{ d} - (1 + 0.15/\text{d} \cdot 2 \text{ d})}$$

$$= 20 \text{ mg/l}\frac{1.3}{16.8 - 1.3}$$

$$= 1.7 \text{ mg/l}$$

Thus, almost all of $S^0$ is removed. Also, $1 + b\theta_x = 1.3$ is another recurring parameter grouping.

Fourth, we determine the effluent (and reactor) concentrations of active, inert, and total volatile solids,

$$X_a = Y(S^0 - S)\frac{1}{1 + b\theta_x}$$
$$= 0.42\frac{\text{g VSS}_a}{\text{g BOD}_L}\left((500 - 1.7)\frac{\text{mg BOD}_L}{1}\right)\frac{1}{1.3}$$
$$= 161 \text{ mg VSS}_a/\text{l}$$
$$X_i = X_i^0 + X_a(1 - f_d)b\theta_x$$
$$= 50 \text{ mg VSS}_i/\text{l} + 161\frac{\text{mg VSS}_a}{1}\left((1 - 0.8)\frac{\text{g VSS}_i}{\text{g VSS}_a}\right) \cdot 0.15/\text{d} \cdot 2 \text{ d}$$
$$= 50 + 9.7 = 59.7 \text{ mg VSS}_i/\text{l} \simeq 60 \text{ mg VSS}_i/\text{l}$$
$$X_v = X_a + X_i = 161 + 60 = 221 \text{ mg VSS/l}$$

These calculations illustrate that the chemostat, while removing nearly all 500 mg $BOD_L$/l of substrate, produces 171 mg VSS/l of biomass and passes through the 50 mg $VSS_i$/l of inert volatile solids.

Fifth, we estimate the effluent SMP. The individual terms required for Equations 3.38 and 3.39 are computed as

$$r_{\text{ut}} = -(S^0 - S)/\theta = -((500 - 1.7) \text{ mg BOD}_L/\text{l})/2 \text{ d} = -249 \text{ mg BOD}_L/\text{l} \cdot \text{d}$$
$$(\hat{q}_{\text{UAP}}X_a\theta + K_{\text{UAP}} + k_1 r_{\text{ut}}\theta) = (1.8 \cdot 161 \cdot 2 + 100 + 0.12 \cdot (-249) \cdot 2)\text{mg COD}_p/\text{l}$$
$$= 620 \text{ mg COD}_p/\text{l}$$
$$-4K_{\text{UAP}}k_1 r_{\text{ut}}\theta = -4 \cdot 100 \cdot 0.12 \cdot (-249) \cdot 2 = 23{,}900 \text{ (mg COD}_p/\text{l)}^2$$
$$K_{\text{BAP}} + (\hat{q}_{\text{BAP}} - k_2)X_a\theta = 85 + (0.1 - 0.09) \cdot 161 \cdot 2 = 88.2 \text{ mg COD}_p/\text{l}$$
$$4K_{\text{BAP}}k_2 X_a\theta = 4 \cdot 85 \cdot 0.09 \cdot 161 \cdot 2 = 9{,}850 \text{ (mg COD}_p/\text{l)}^2$$

Then, UAP and BAP are

$$\text{UAP} = \frac{-620 + \sqrt{(620)^2 + 23{,}900}}{2}$$
$$= 9.5 \text{ mg COD}_p/\text{l}$$

$$\text{BAP} = \frac{-88.2 + \sqrt{(88.2)^2 + 9{,}850}}{2}$$
$$= 22.3 \text{ mg COD}_p/\text{l}$$

$$\text{SMP} = \text{UAP} + \text{BAP} = 9.5 + 22.3 = 31.8 \text{ mg COD}_p/\text{l}$$

Sixth, we compute the total effluent quality. The soluble COD is equal to the sum of substrate and SMP

$$\text{Soluble COD} = S + \text{SMP}$$
$$= 1.7 + 31.8$$
$$= 33.5 \text{ mg COD/l}$$

As is often the case, the soluble COD is dominated by SMP, not original substrate. The soluble

$BOD_L$ will equal the soluble COD if all of the SMP is biodegradable, which we assume is true:

$$\text{Soluble } BOD_L = 33.5 \text{ mg } BOD_L/\text{l}$$

The total effluent COD is the sum of the soluble COD and the COD of all the volatile solids.

$$\begin{aligned}\text{Total COD} &= \text{Soluble COD} + (1.42 \text{ g COD/gVSS}) \cdot X_v \\ &= 33.5 + 1.42 \cdot 221 \\ &= 347 \text{ mg COD/l.}\end{aligned}$$

The total $BOD_L$ is the sum of the soluble $BOD_L$ (we assume that the SMP is degradable) and the oxygen demand of the biodegradable fraction of the *active* biomass.

$$\begin{aligned}\text{Total } BOD_L &= \text{Soluble } BOD_L + (1.42 \text{ g BOD/g } VSS_a) \cdot X_a \cdot f_d \\ &= 33.5 + 1.42 \cdot 161 \cdot 0.8 \\ &= 216 \text{ mg } BOD_L/\text{l}\end{aligned}$$

We see that most of the effluent COD and $BOD_L$ in a chemostat come from the volatile suspended solids. Also, the total COD and $BOD_L$ computations utilize the conversion factor 1.42 g $BOD_L$ (or COD)/gVSS, which comes from the oxygen demand of C in $C_5H_7O_2N$.

**Example 3.2**

**DESIGN OF A CHEMOSTAT** Design requires that certain effluent criteria be met. This example proposes several types of design criteria and shows how the choice of the "best" design $\theta_x$ depends on the criterion used. The influent flow and quality remain the same as in Example 3.1, but $V$ can be altered to control $\theta_x$. The design problem is solved by first creating for a set of $\theta_x$ values (from 0.2 to 100 d) the effluent concentrations for $S$, $X_a$, $X_v$, SMP, soluble

**Table 3.2** Summary of effluent quality for the parameters of Example 3.1 and the range of $\theta_x$ values for Example 3.2

| | | | | | $BOD_L$, mg $BOD_L$/l | |
|---|---|---|---|---|---|---|
| $\theta_x$, d | $S$, mg $BOD_L$/l | $X_a$, mg $VSS_a$/l | $X_v$, mg VSS/l | SMP, mg $COD_p$/l | Soluble | Total |
| 0.20 | 31.7 | 191 | 242 | 40.8 | 72.4 | 289 |
| 0.25 | 19.5 | 195 | 246 | 39.1 | 58.7 | 280 |
| 0.5 | 6.9 | 193 | 246 | 32.6 | 39.5 | 258 |
| 1 | 3.2 | 181 | 236 | 29.4 | 32.6 | 239 |
| 2 | 1.7 | 161 | 221 | 31.8 | 33.5 | 216 |
| 3 | 1.2 | 144 | 207 | 35.4 | 36.7 | 201 |
| 5 | 0.87 | 120 | 188 | 41.4 | 42.3 | 178 |
| 10 | 0.61 | 84 | 159 | 49.9 | 50.5 | 146 |
| 15 | 0.53 | 65 | 144 | 54.2 | 54.8 | 126 |
| 20 | 0.49 | 52 | 134 | 56.9 | 57.4 | 117 |
| 30 | 0.45 | 38 | 122 | 59.9 | 60.3 | 104 |
| 50 | 0.41 | 25 | 112 | 62.7 | 63.1 | 91 |
| 100 | 0.39 | 13 | 102 | 65.0 | 65.4 | 80 |

I Recall that $S_{min}$ = 0.36 mg $BOD_L$/l and $\theta_x^{min}$ = 0.125 d.

**Table 3.3** Selection of the most appropriate $\theta_x$ value (from Table 3.2) for different design criteria

| Criteria | Design $\theta_x$ |
|---|---|
| Reduce original substrate to 1 mg/l or less | $\theta_x >\sim 4$ d |
| 95% removal of original substrate | $\theta_x >\sim 0.22$ d |
| 99% removal of original substrate | $\theta_x >\sim 0.7$ d |
| Minimize soluble $BOD_L$ | $\theta_x \sim 1$ d |
| 95% removal of soluble $BOD_L$ | not feasible |
| 90% removal of soluble $BOD_L$ | $0.4 < \theta_x < 9$ d |
| Minimize total $BOD_L$ | $\theta_x$ large |
| Maximize the generation of VSS | $\theta_x$ from 0.25–0.5 d |
| Minimize the generation of SMP | $\theta_x \sim 1$ d |
| Maximize the fraction of VSS that is active | $\theta_x \sim 0.25$ d |

$BOD_L$, and total $BOD_L$. Each is computed in the same manner as was done in Example 3.1. Table 3.2 summarizes the computations.

Table 3.3 summarizes how different design criteria use the results to choose the most appropriate $\theta_x$. These results show that $\theta_x$ chosen from effluent $BOD_L$ criteria are not controlled by original substrate alone. Instead, SMP and $VSS_a$ are more important than $S$ when real-world effluent criteria are used.

**Example 3.3** **EFFLUENT $BOD_5$** The results in Tables 3.2 and 3.3 use $BOD_L$ as the BOD criterion, not $BOD_5$, which is most often used in standards and regulations. It is quite likely that the ratios of $BOD_5$:$BOD_L$ are very different for original substrate, SMP, and active biomass. Although much needs to be learned about the exertion of BOD by the different sources, this example shows how conversions can be made.

The BOD exertion equation is

$$BOD_t = BOD_L(1 - \exp\{-kt\}) \tag{3.41}$$

in which $BOD_t$ is the BOD exerted (i.e., measured by dissolved-oxygen uptake) at time $t$, $k$ is the first-order rate constant for BOD exertion ($T^{-1}$), and exp is the exponential function. From Equation 3.41, the ratio $BOD_5$:$BOD_L$ is

$$BOD_5{:}BOD_L = 1 - \exp\{-k \cdot 5\ \mathrm{d}\} \tag{3.42}$$

We shall assume that $k = 0.23$/d for original substrate, 0.1/d for active biomass, and 0.03/d for SMP. These three values are based, respectively, on the typical BOD-exertion kinetics for readily biodegraded substrates, decay rate of active biomass, and BAP-utilization kinetics. While not universal, they give a good first estimate of the relative biodegradability of each of the three components of the effluent $BOD_L$. The $BOD_5$:$BOD_L$ ratios are then 0.68 for original substrate, 0.40 for active biomass, and 0.14 for SMP.

These ratios are combined with the data in Table 3.2 to compute the weight-averaged soluble and total $BOD_5$ values shown in Table 3.4. When soluble $BOD_5$ is used, the minimum value occurs for an SRT of 2 to 3 d, instead of 1 d (Table 3.3). Furthermore, the soluble $BOD_5$

**Table 3.4** Estimated weight-averaged $BOD_5$ concentrations for the results in Table 3.2

| $\theta_x$, d | Soluble $BOD_5$, mg/l | Total $BOD_5$, mg/l |
|---|---|---|
| 0.20 | 27.3 | 114 |
| 0.25 | 18.8 | 107 |
| 0.50 | 9.2 | 97 |
| 1 | 6.3 | 89 |
| 2 | 5.6 | 79 |
| 3 | 5.8 | 71 |
| 5 | 6.4 | 61 |
| 10 | 7.4 | 46 |
| 15 | 8.0 | 37 |
| 20 | 8.3 | 32 |
| 30 | 8.7 | 26 |
| 50 | 9.1 | 20 |
| 100 | 9.4 | 15 |

concentrations are close to those observed in practical aerobic systems, in which soluble $BOD_5$ normally is less than 10 mg/l and is relatively insensitive to changes in $\theta_x$ within its usual range (2 to 50 d).

## 3.6 NUTRIENTS AND ELECTRON ACCEPTORS

A process in environmental biotechnology must provide sufficient nutrients and electron acceptors to support biomass growth and energy generation. Nutrients, being elements comprising the physical structure of the cells, are needed in proportion to the net production of biomass. Active and inert biomass contain nutrients, as long as they are produced microbiologically. The electron acceptor is consumed in proportion to electron-donor utilization multiplied by the sum of exogenous and endogenous flows of electron to the terminal acceptor.

Nutrient and acceptor requirements can be determined from the stoichiometric equations developed in Chapter 2. The net yield fraction, $f_s$, is used to construct the overall reaction including synthesis and endogenous decay. Equation 3.33 relates $f_s$ to $f_s^0$ and $\theta_x$.

Another approach obtains nutrient requirements directly from the reactor mass balances. This approach is advantageous when materials other than biomass and substrate are important. SMP are key examples of other important materials. In terms of our chemostat model, the rate of nutrient consumption is

$$r_n = \gamma_n Y r_{\text{ut}} \frac{1 + (1 - f_d) b\theta_x}{1 + b\theta_x} \quad \textbf{[3.43]}$$

in which

$r_n$ = rate of nutrient consumption ($M_n L^{-3} T^{-1}$)
$\gamma_n$ = the stoichiometric ratio of nutrient mass to VSS for the biomass [$M_n M_x^{-1}$].

Recall that $r_{ut}$ is the rate of substrate utilization and has a negative sense (Equation 3.40), which also makes $r_n$ negative. The most important nutrients are $N$ and $P$. The empirical formula for bacterial VSS, $C_5H_7O_2N$, has $\gamma_N$ = 14 g N/113 g VSS = 0.124 g N/g VSS. Generally, the $P$ requirement is 20 percent of the $N$ requirement, which makes $\gamma_P$ = 0.025 g P/g VSS. An overall mass balance on a nutrient is then

$$0 = QC_n^0 - QC_n + r_n V \quad \textbf{[3.44]}$$

in which $C_n$ and $C_n^0$ are the effluent and influent nutrient concentrations ($M_n L^{-3}$), respectively. Solution for $C_n$ gives

$$C_n = C_n^0 + r_n \theta \quad \textbf{[3.45]}$$

The supply of nutrients must be supplemented if $C_n$ takes a negative value in Equation 3.45.

A parallel means to compute the use rate of electron acceptor (denoted $\Delta S_a/\Delta t$ and having units $M_a T^{-1}$) is by a mass balance on electron equivalents expressed as oxygen demand. For the chemostat, the input of oxygen equivalents or oxygen demand (OD) includes the electron-donor substrate and the input VSS:

$$\text{OD inputs} = QS^0 + 1.42 \frac{\text{g COD}}{\text{g VSS}} X_v^0 Q \quad \textbf{[3.46]}$$

The outputs are residual substrate, SMP, and all VSS:

$$\text{OD outputs} = QS + Q(\text{SMP}) + 1.42 \frac{\text{g COD}}{\text{g VSS}} X_v Q \quad \textbf{[3.47]}$$

The acceptor consumption as $O_2$ equivalents is the difference between the inputs and the outputs.

$$\frac{\Delta S_a}{\Delta t} = \gamma_a Q \left[ S^0 - S - \text{ SMP } + 1.42(X_v^0 - X_v) \right] \quad \textbf{[3.48]}$$

in which $\gamma_a$ is the stoichiometric ratio of acceptor mass to oxygen demand. For oxygen, $\gamma_a$ = 1 g $O_2$/g COD; for $NO_3^-$-N, $\gamma_a$ is 0.35 g $NO_3^-$-N/g COD. Other $\gamma_a$ ratios can be computed from stoichiometry of the electron-acceptor half-reactions presented in Chapter 2.

The acceptor can be supplied in the influent flow or by other means, such as aeration to provide oxygen. This can be expressed as

$$\frac{\Delta S_a}{\Delta t} = Q \left[ S_a^0 - S_a \right] + R_a \quad \textbf{[3.49]}$$

in which $S_a$ and $S_a^0$ are the effluent and influent concentrations of the acceptor, and $R_a$ is the required mass rate of acceptor supply ($M_a T^{-1}$).

**Example 3.4**

**NUTRIENT AND ACCEPTOR CONSUMPTION** For Example 3.1, we computed the following:

$$
\begin{aligned}
S^0 &= 500 \text{ mg BOD}_L/\text{l} \\
S &= 1.7 \text{ mg BOD}_L/\text{l} \\
X_i^0 &= 50 \text{ mg VSS}_i/\text{l} \\
X_v &= 221 \text{ mg VSS/l} \\
\text{SMP} &= 31.8 \text{ mg BOD}_L/\text{l} \\
r_{\text{ut}} &= -249 \text{ mg BOD}_L/\text{l}.
\end{aligned}
$$

In this example, the input concentrations of N, P, and $O_2$ are 50 mg $NH_4^+$-N/l, 10 mg $PO_4^{3-}$-P/l, and 6 mg $O_2$/l, respectively.

The N and P consumption rates are

$$
\begin{aligned}
r_N &= 0.124 \frac{\text{g N}}{\text{g VSS}} \cdot 0.42 \frac{\text{g VSS}}{\text{g BOD}_L} \cdot \left(-249 \frac{\text{mg BOD}_L}{\text{l-d}}\right) \cdot \frac{1 + (1 - 0.8) \cdot 0.15 \cdot 2}{1 + 0.15 \cdot 2} \\
&= -10.6 \text{ mg N/l-d}
\end{aligned}
$$

$$
r_P = r_N \cdot 0.2 \frac{\text{g P}}{\text{g N}} = -10.6 \cdot 0.2 = -2.1 \text{ mg P/l-d}
$$

The mass balances give the effluent concentrations

$$
\begin{aligned}
C_N &= 50 \text{ mg N/l} - 10.6 \frac{\text{mg N}}{\text{l-d}} \cdot 2 \text{ d} \\
&= 28.8 \text{ mg NH}_4^+\text{-N/l} \\
C_P &= 10 \text{ mg P/l } - 2.1 \frac{\text{mg P}}{\text{l-d}} \cdot 2 \text{ d} \\
&= 5.8 \text{ mg PO}_4^{3-}\text{-P/l}
\end{aligned}
$$

In both cases, sufficient nutrients are contained in the influent.

For the electron acceptor, $O_2$ in this case, the required overall supply rate is

$$
\begin{aligned}
\frac{\Delta S_a}{\Delta t} &= 1 \frac{\text{g O}_2}{\text{g COD}} \cdot \left(1{,}000 \frac{\text{m}^3}{\text{d}}\right) \\
&\quad \cdot [500 - 1.7 - 31.8 + 1.42(50 - 221)] \text{ mg COD/l} \cdot \frac{10^3 \text{l}}{\text{m}^3} \cdot 10^{-3} \frac{\text{g}}{\text{mg}} \\
&= 2.24 \cdot 10^5 \text{ g O}_2/\text{d}.
\end{aligned}
$$

Clearly, the influent concentration of 6 mg/l is not going to supply the $O_2$ required. Therefore, oxygen must be supplied at a rate ($R_{O_2}$) of

$$
R_{O_2} = \frac{\Delta S_{O_2}}{\Delta t} - Q[S_a^0 - S_a]
$$

For an aerobic system, $S_a$ is typically 2 mg/l. Therefore,

$$
\begin{aligned}
R_{O_2} &= 2.24 \cdot 10^5 \frac{\text{g O}_2}{\text{d}} - [1{,}000 \text{m}^3/\text{d}][6 \text{mg/l} - 2 \text{ mg/l}] \cdot \frac{10^3 \text{ l}}{\text{m}^3} \cdot \frac{10^{-3} \text{ g}}{\text{mg}} \\
&= 2.20 \cdot 10^5 \text{ g O}_2/\text{d}
\end{aligned}
$$

If $R_{O_2}$ is divided by $Q$, the rate of acceptor supply is expressed as a concentration in the influent flow: Here it is 220 g $O_2/m^3$ = 220 mg $O_2$/l. Clearly, normal oxygen in the influent cannot supply the needed oxygen, and aeration is essential.

## 3.7 INPUT ACTIVE BIOMASS

Whether by design or by happenstance, some biological processes receive significant inputs of biomass active in degradation of the substrate. Three circumstances provide practical examples. First, when microbial processes are operated in series, the downstream process often receives significant biomass from the upstream process. Second, microorganisms may be discharged in a waste stream or grown in the sewers. Third, bioaugmentation is the deliberate addition of microorganisms to improve some aspect of process performance.

When active biomass is input, the steady-state mass balance for active biomass, Equation 3.16, must be modified.

$$0 = QX_a^0 - QX_a + Y\frac{\hat{q}S}{K+S}X_aV - bX_aV \quad \textbf{[3.50]}$$

The other mass balances remain the same. Equation 3.50 can be solved for $S$ when the SRT is redefined mathematically to maintain its definition as the reciprocal of the net specific growth rate:

$$S = K\frac{1+b\theta_x}{Y\hat{q}\theta_x - (1+b\theta_x)} \quad \textbf{[3.51]}$$

where

$$\theta_x = \mu^{-1} = \frac{X_aV}{QX_a - QX_a^0} \quad \textbf{[3.52]}$$

The solution for $S$ is exactly the same as it was for the chemostat before (Equation 3.24), but $\theta_x$ is now defined with the denominator being the gross active biomass output rate ($QX_a$) minus the input rate ($QX_a^0$). Thus, the denominator remains the net production rate of new active biomass.

Figure 3.5 illustrates how $X_a^0$ affects $S$ and washout. In the example, $S^0$ is fixed at 100 mg/l, and the parameter values used are $Y = 0.44$ mg VSS$_a$/mg, $\hat{q} = 5$ mg/mg VSS$_a$-d, $K = 20$ mg/l, and $b = 0.2$/d. Without input active biomass, washout occurs for $\theta_x$ of about 0.6 d. As $X_a^0$ increases, complete washout is eliminated, because the reactor always contains some biomass. Increasing $X_a^0$ also makes the substrate concentration lower, and the effect is most dramatic near washout.

Although the mass balances for substrate, inert, and volatile biomass remain the same as before, their solutions differ, due to the change in definition of $\theta_x$. In particular, mass balance Equations 3.17 and 3.50, together with Equation 3.52, give

$$X_a = \frac{\theta_x}{\theta}\left[Y(S^0 - S)\frac{1}{1+b\theta_x}\right] \quad \textbf{[3.53]}$$

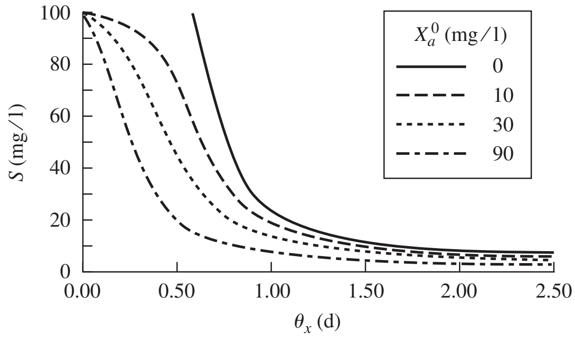


**Figure 3.5** Effect of influent active biomass and $\theta_x$ on effluent substrate concentration for a chemostat at steady state. The parameters used are $Y = 0.44$ mg VSS$_a$/mg, $\hat{q} = 5$ mg/mg VSS$_a$-d, $K = 20$ mg/l, and $b = 0.2$/d.

Equation 3.29 then yields

$$X_i = X_i^0 + (1 - f_d)bX_a\theta \tag{3.54}$$

$$X_v = X_i + X_a = X_i^0 + (1 + (1 - f_d)b\theta)X_a \tag{3.55}$$

The concentration of active biomass is "built up" by input of active biomass, and this "build up" is quantified by the ratio $\theta_x/\theta$. Rearrangement of Equation 3.52 shows that the $\theta_x/\theta$ ratio depends on the relative concentrations of $X_a$ and $X_a^0$.

$$\frac{\theta_x}{\theta} = \frac{1}{1 - X_a^0/X_a} \tag{3.56}$$

When $X_a^0$ approaches, but is less than $X_a$, $\theta_x$ is substantially greater than $\theta$. It is even possible to have $X_a^0 > X_a$, if $X_a^0$ is very large. In such a case, $\theta_x$ is negative, and the process is in net biomass decay. Supplying a very large $X_a^0$ is a means to make the steady-state $\mu$ negative and sustain $S < S_{min}$.

Once $S$ and $X_a$ are determined from the proper definition of $\theta_x$, UAP and BAP concentrations are computed from Equations 3.38 and 3.39, as usual.

**Example 3.5**

**INPUT ACTIVE BIOMASS** We recompute the results of Examples 3.1 and 3.4 if the system also receives input active biomass at a concentration of $X_a^0 = 40$ mg VSS$_a$/l. To do the computations, Equations 3.51, 3.53, and 3.56 must be solved iteratively. The strategy is to pick a $\theta_x \geq \theta$, compute $S$ from Equation 3.51, compute $X_a$ from Equation 3.53, compute $\theta_x$ from Equation 3.56, and compare the computed $\theta_x$ to the original $\theta_x$. If they do not agree, pick another $\theta_x$ and iterate. Applying that strategy yields:

| | |
|---|---|
| $\theta_x = 2.53$ d | (up from 2.0 with no $X_a^0$) |
| $S = 1.4$ mg BOD$_L$/l | (down from 1.7) |
| $X_a = 191$ mg VSS$_a$/l | (up from 161) |

Further solution with Equations 3.54 and 3.55 gives

$$X_i = 61 \text{ mg VSS}_i/\text{l} \qquad \text{(up from 60)}$$
$$X_v = 252 \text{ mg VSS/l} \qquad \text{(up from 221)}$$

Thus, addition of 40 mg VSS$_a$/l increases $X_a$ by 30 mg VSS$_a$/l and $X_v$ by 31 mg VSS/l, while $\theta_x$ goes up by 26.5 percent. The increase in $\theta_x$ means that biomass decay increases, and $X_a$ increases by less than $X_a^0$. These are the intuitively expected effects of adding active biomass.

Computation of UAP and BAP is altered only by the changes in $r_{ut}$ and $X_a$.

$$r_{ut} = -(500 - 1.4 \text{ mg BOD}_L/\text{l})/2 \text{ d}$$
$$= -249 \text{ mg BOD}_L/\text{l} \qquad \text{(not changed perceptively)}$$
$$\text{UAP} = 8.1 \text{ mg COD}_p/\text{l} \qquad \text{(down from 9.5)}$$
$$\text{BAP} = 25.6 \text{ mg COD}_p/\text{l} \qquad \text{(up from 22.3)}$$
$$\text{SMP} = 33.7 \text{ mg COD}_p/\text{l} \qquad \text{(up from 31.8)}$$

The increased active biomass causes greater BAP formation and accumulation of SMP.

For the total effluent quality,

$$\text{Soluble BOD}_L \text{ or COD} = 1.4 + 33.7$$
$$= 35.1 \text{ mg BOD}_L/\text{l} \qquad \text{(up from 33.5)}$$
$$\text{Total COD} = 35.1 + 1.42 \cdot 252$$
$$= 393 \text{ mg COD/l} \qquad \text{(up from 348)}$$
$$\text{Total BOD}_L = 35.1 + 1.42 \cdot 191 \cdot 0.8$$
$$= 252 \text{ mg BOD}_L/\text{l} \qquad \text{(up from 216)}$$

So, although adding active biomass increases $X_a$ and $\theta_x$ and decreases $S$, the effluent quality has higher soluble and total BOD$_L$ and COD. This is the result of adding active biomass.

The oxygen requirement is

$$\frac{\Delta S_{O_2}}{\Delta t} = 1 \cdot 1{,}000 \cdot [500 - 1.4 - 33.7 + 1.42(90 - 252)]$$
$$= 2.35 \cdot 10^5 \text{ g O}_2/\text{d} \qquad (\text{up from } 2.24 \cdot 10^5)$$

Thus, the endogenous decay of the added biomass slightly increases the oxygen required.

Nutrient requirements are

$$r_N = 0.124 \cdot 0.42 \cdot (-249) \frac{1 + (1 - 0.8) \cdot 0.15 \cdot 2.53}{1 + 0.15 \cdot 2.53}$$
$$= -10.1 \text{ mg N/l-d} \qquad (\text{down from } -10.6)$$
$$r_P = -10.1 \cdot 0.2 = -2.0 \text{ mg P/l-d } (\text{down from } -2.1)$$

Thus, the net synthesis of biomass is less and requires less $N$ and $P$ as nutrients.

## 3.8 Hydrolysis of Particulate and Polymeric Substrates

Several important applications in environmental biotechnology involve organic substrates that originate as particles or large polymers. For example, more than one-half of the BOD in typical sewage is comprised of suspended solids, and sludges undergoing digestion are nearly 100 percent particulate BOD. Before the bacteria can begin the oxidation reactions characteristic of the catabolism of organic substrates, the particles or large polymers must be hydrolyzed to smaller molecules that can be transported across the cell membrane. Extracellular enzymes catalyze these hydrolysis reactions.

Despite its great importance in many situations, hydrolysis has not been thoroughly researched. The best way to represent hydrolysis kinetics is not settled. Part of the problem comes about because the hydrolytic enzymes are not necessarily associated with or proportional to the active biomass, although the active biomass produces them. Exactly what controls the level of hydrolytic enzymes is not established, and their measurement is neither simple, nor has it been carried out frequently in systems relevant to environmental biotechnology.

A simple, but reasonably reliable approach for describing hydrolysis kinetics is a first-order relationship with respect to the particulate (or large-polymer) substrate:

$$r_{\text{hyd}} = -k_{\text{hyd}} S_p \qquad \textbf{[3.57]}$$

in which

$r_{\text{hyd}}$ = rate of accumulation of particulate substrate due to hydrolysis ($M_sL^{-3}T^{-1}$)

$S_p$ = concentration of the particulate (or large-polymer) substrate ($M_sL^{-3}$)

$k_{\text{hyd}}$ = first-order hydrolysis rate coefficient ($T^{-1}$)

In principle, $k_{\text{hyd}}$ is proportional to the concentration of hydrolytic enzymes, as well as to the intrinsic hydrolysis kinetics of the enzymes. Some workers include the active biomass concentration as part of $k_{\text{hyd}}$ (i.e., $k_{\text{hyd}} = k'_{\text{hyd}} X_a$, in which $k'_{\text{hyd}}$ is a specific hydrolysis rate coefficient ($L^3M_x^{-1}T^{-1}$). An advantage of using $k'_{\text{hyd}} X_a$ is that the hydrolysis rate automatically drops to zero when biomass is not present. On the other hand, it also implies that the extracellular enzymes are linearly proportional to the biomass, an assertion that is not proven.

When Equation 3.57 is used for the hydrolysis rate, the steady-state mass balance on particulate substrate in a chemostat is

$$0 = Q(S_p^0 - S_p) - k_{\text{hyd}} S_p V \qquad \textbf{[3.58]}$$

in which $S_p^0$ = the effluent concentration of particulate substrate ($M_sL^{-3}$). Solving Equation 3.58 gives

$$S_p = \frac{S_p^0}{1 + k_{\text{hyd}}\theta} \qquad \textbf{[3.59]}$$

Here, $\theta$ represents the liquid detention time and should not be substituted by $\theta_x$.

The destruction of particulate substrate results in the formation of soluble substrate with conservation of the electron equivalents, or $BOD_L$. When both substrate types have the same mass measure (generally oxygen demand as a surrogate for electron equivalents), the formation rate of soluble substrate is simply $k_{\text{hyd}} S_p$. Then, the steady-state mass balance on soluble substrate is

$$0 = (S^0 - S) - \frac{\hat{q}S}{K + S} X_a V/Q + k_{hyd} S_p V/Q \qquad [3.60]$$

Because Equation 3.60 shows an additional source of substrate from hydrolysis, $S^0$ effectively is increased by $k_{hyd} S_p V/Q$; the amount of biomass accumulated should be augmented.

Other constituents of particulate substrates also are conserved during hydrolysis. Good examples are the nutrients nitrogen, phosphorus, and sulfur. The formation rate for soluble forms of these nutrients is

$$r_{hydn} = \gamma_n k_{hyd} S_p \qquad [3.61]$$

in which

$r_{hydn}$ = rate of accumulation of a soluble form of nutrient $n$ by hydrolysis ($M_n L^{-3} T^{-1}$)

$\gamma_n$ = stoichiometric ratio of nutrient $n$ in the particulate substrate ($M_n M_s^{-1}$).

**Example 3.6** **EFFECT OF HYDROLYSIS** Example 3.1 showed that a chemostat fed 500 mg $BOD_L$/l with a liquid detention time of 2 d produced an effluent quality of:

$$S = 1.7 \text{ mg BOD}_L/\text{l}$$
$$X_a = 161 \text{ mg VSS}_a/\text{l}$$
$$X_v = 221 \text{ mg VSS/l}$$
$$\text{SMP} = 32 \text{ mg COD}_p/\text{l}$$
$$\text{Soluble BOD}_L = 33.5 \text{ mg BOD}_L/\text{l}$$
$$\text{Total COD} = 348 \text{ mg COD/l}$$

We consider now that the influent also contains biodegradable particulate organic matter with a concentration of 100 mg COD/l, and the hydrolysis rate coefficient is $k_{hyd} = 3$/d.

The computations to predict the new effluent quality proceed with the following steps:

1. $S_p$ is computed from Equation 3.59:

$$S_p = \frac{100 \text{ mg COD/l}}{1 + (3/\text{d})(2\text{ d})}$$
$$= 14 \text{ mg COD/l}$$

2. Since $\theta_x = \theta$ remains 2 d, and no active biomass is input, $S = 1.7$ mg $BOD_L$/l.
3. The effluent and reactor biomass concentrations are determined in parallel to example 3.1, except that the effective $S^0$ is now:

$$S^0 = 500 \text{ mg/l} + (100 - 14)\text{mg/l}$$
$$= 586 \text{ mg BOD}_L/\text{l}$$

$$X_a = 0.42 \frac{\text{g VSS}_a}{\text{g BOD}_L} \left( (586 - 1.7) \frac{\text{mg BOD}_L}{1} \right) \frac{1}{1.3}$$
$$= 189 \text{ mg VSS}_a/\text{l}$$

$$X_i = 50 \text{ mg VSS}_i/\text{l} + 189 \frac{\text{mg VSS}_a}{1} (1 - 0.8)(0.15/\text{d})(2\text{ d})$$
$$= 61 \text{ mg VSS}_i/\text{l};$$

$$X_v = X_a + X_i + S_p = 189 + 61 + \frac{14}{1.42} = 264 \text{ mg VSS/l}$$

Thus, the amount of active biomass is augmented by the hydrolysis of particulate COD, while the VSS also is augmented by the remaining biodegradable particulate COD.

**4.** The detailed computations for SMP are omitted, as they are exactly analogous to Example 3.1. The result is

$$\text{UAP} = 9.8 \text{ mg COD}_p/\text{l}$$
$$\text{BAP} = 25.3 \text{ mg COD}_p/\text{l}$$
$$\text{SMP} = 35.1 \text{ mg COD}_p/\text{l}$$

**5.** The effluent concentrations of COD and $\text{BOD}_L$ are affected by the changes in $X_a$, $X_i$ and SMP, as well as by the residual particulate organic substrate $S_p$. All increase.

$$\begin{aligned}
\text{Soluble COD and BOD}_L &= S + \text{SMP} \\
&= 1.7 + 35.1 \\
&= 36.8 \text{ mg COD/l} \\
\text{Total COD} &= S + \text{SMP} + 1.42 \cdot X_v \\
&= 1.7 + 35.4 + 1.42 \cdot 264 \\
&= 412 \text{ mg COD/l} \\
\text{Total BOD}_L &= S + \text{SMP} + 1.42 \cdot f_d \cdot X_a + S_p \\
&= 1.7 + 35.4 + 1.42 \cdot 0.8 \cdot 189 + 14 \\
&= 266 \text{ mg BOD}_L/\text{l}
\end{aligned}$$

## 3.9 INHIBITION

The rates of substrate utilization and microbial growth can be slowed by the presence of inhibitory compounds. Examples of inhibitors are heavy metals, pesticides, antibiotics, aromatic hydrocarbons, and chlorinated solvents. Sometimes these materials are called toxicants, and their effects termed toxicity. Here, we use the terms inhibitor and inhibition, because they imply one of several different general phenomena affecting metabolic rates.

The range of possible inhibitors and their different effects on the microorganisms can make inhibition a confusing topic. In some cases, the inhibitor affects a single enzyme active in substrate utilization; in such a case, utilization of that substrate is slowed. In other cases, the inhibitor affects some more general cell function, such as respiration; then, indirect effects, such as reduced biomass levels may slow utilization of a particular substrate. Finally, some reactions actually are increased by inhibition, as the cell tries to compensate for the negative impacts of the inhibitor.

Figure 3.6 shows the key places that inhibitors affect the primary flows of electrons and energy. Inhibition of a particular degradative enzyme may occur during the initial oxidation reactions of an electron-donor substrate. The immediate effect is a slowing of the degradation rate. In addition, the reduced electron flow can lead to a loss of biomass or slowing of other reactions requiring electrons ($ICH_2$) or energy (ATP). At the other end of the electron-transport chain, inhibition of the acceptor reaction prevents electron flow and energy generation, thereby leading to a loss of biomass. Interestingly, inhibition of the acceptor reaction can lead to a buildup of reduced electron carriers ($ICH_2$) and increased rates for reactions requiring reduced electron carriers as a cosubstrate (Wrenn and Rittmann, 1995). Decouplers reduce or eliminate energy generation, even though electrons flow from the donors to the acceptor. Decoupling inhibits cell growth and other energy-requiring reactions. In some cases, decoupling inhibition leads to an increase in acceptor utilization per unit biomass, as the cells attempt to compensate for a low energy yield by sending more electrons to the acceptor.

How an inhibitor affects growth and substrate-utilization kinetics can be expressed succinctly by using *effective kinetic parameters*. The kinetic expressions for substrate utilization and growth remain the same as before (i.e., Equations 3.6 and 3.9), but the effective kinetic parameters depend on the concentration of the inhibitor. Written out with effective parameters, Equations 3.6 and 3.9 are

$$r_{\text{ut, eff}} = -\frac{\hat{q}_{\text{eff}} S}{K_{\text{eff}} + S} X_a \quad \textbf{[3.62]}$$

$$\mu_{\text{eff}} = Y_{\text{eff}}(-r_{\text{ut, eff}}) - b_{\text{eff}} \quad \textbf{[3.63]}$$

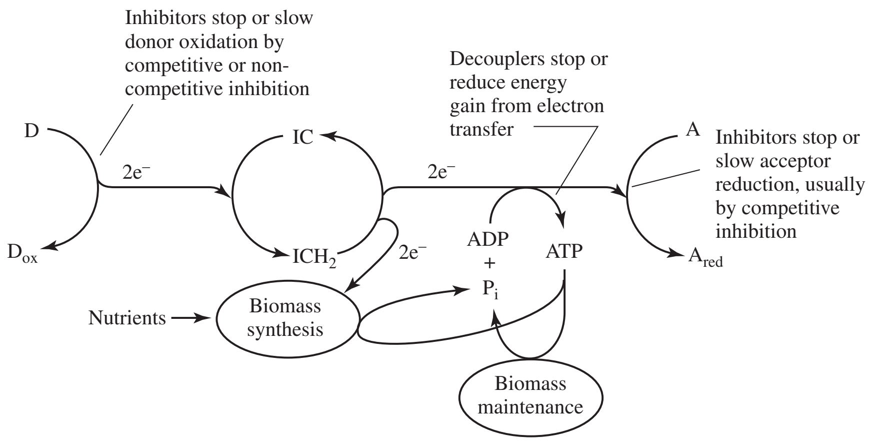


**Figure 3.6** Illustration of how inhibitions can affect the primary flow of electrons and energy. *D* = electron donor, *A* = electron acceptor, and IC = intracellular electron carrier, such as $NAD^+$.
(After Rittmann and Sáez, 1993).

How $\hat{q}_{\text{eff}}$, $K_{\text{eff}}$, $Y_{\text{eff}}$, and $b_{\text{eff}}$ are controlled by the inhibitor concentrations depends on the location and mode of the inhibition phenomenon. The most common types of inhibition and their effective parameters are reviewed here. Some aspects of inhibition were introduced in Chapter 1, and more details can be found in Rittmann and Sáez (1993).

A common type of inhibition for aromatic hydrocarbons and chlorinated solvents is *self-inhibition,* which also is called *Haldane* or *Andrews kinetics*. In this case, the enzyme-catalyzed degradation of the substrate is slowed by high concentrations of the substrate itself. It is not clear whether the self-inhibition occurs directly through action on the degradative enzyme or indirectly through hindering electrons or energy flow after the original donor reaction. In either situation, the effective parameters for self-inhibition are

$$\hat{q}_{\text{eff}} = \frac{\hat{q}}{1 + \dfrac{S}{K_{\text{IS}}}} \tag{3.64}$$

$$K_{\text{eff}} = \frac{K}{1 + S/K_{\text{IS}}} \tag{3.65}$$

where $K_{\text{IS}}$ = an inhibition concentration of the self-inhibitory substrate ($\text{M}_s\text{L}^{-3}$). $Y_{\text{eff}}$ and $b_{\text{eff}}$ are not affected and remain $Y$ and $b$, respectively.

Figure 3.7 contains two graphical presentations of the effect of substrate self-inhibition on reaction kinetics. The left figure illustrates how the reaction rate ($-r_{\text{ut}}$) varies with substrate concentration. At low substrate concentrations, the rate increases with an increase in $S$. However, a maximum rate is reached, and substrate concentrations beyond this become inhibitory, causing a decrease in reaction rate. When the Haldane reaction rate model is substituted for the Monod relation in the CSTR mass balances, the effect of $\theta_x$ on effluent substrate concentration is illustrated by the right graph in Figure 3.7. This graph indicates that, with $\theta_x$ greater than about 2 d, two values of $S$ are possible for each value of $\theta_x$. Which is the correct one? The answer depends on how the reactor arrives at the steady state.

First, we assume that the influent concentration is 45 mg/l, and that the reactor is being operated at a $\theta_x$ of 10 d, conditions indicated by the dotted lines in the figure. Under such conditions, the influent concentration is quite inhibitory to the microorganisms. If the reactor were initially filled with the untreated wastewater, seeded with microorganisms, and then operated at the $\theta_x$ of 10 d, the reactor would fail, because the specific growth rate of the organisms under these conditions is less than 0.1/d (= $1/\theta_x$). However, if the wastewater were initially diluted so that the concentration in the reactor was about 33 mg/l, then the specific growth rate would be greater than 0.1/d, and the bacteria would reproduce faster than they are removed from the reactor. The reactor population would continue to increase, and the value of $S$ in the reactor would continue to decrease until the microbial population size and substrate concentration reached their steady-state concentrations, about 1 mg/l for the substrate. The value of Figure 3.7, then, is not only to indicate what this steady-state concentration would be for a well-operating reactor, but also to indicate upper limits on substrate concentration beyond which reactor startup cannot proceed.

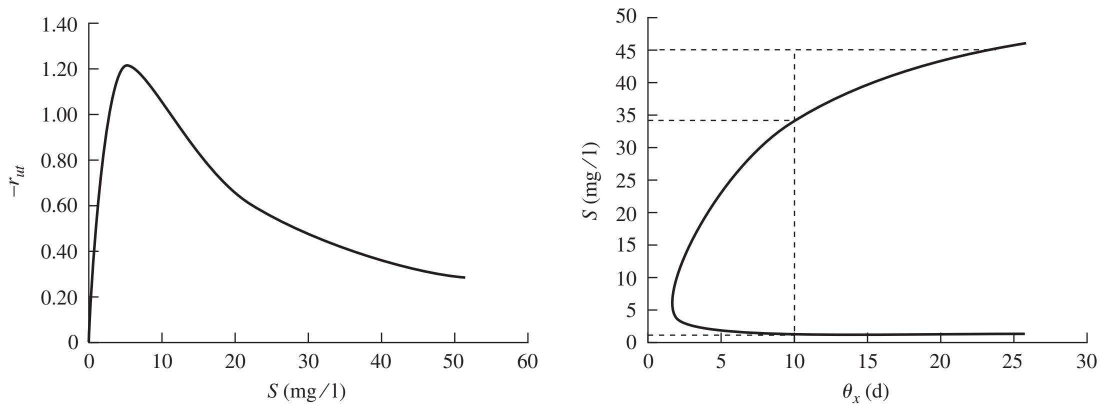


**Figure 3.7** Haldane kinetics and the relationship between $S$ and reaction rate (top) and between $S$ and $\theta_x$. The example is for orthochlorophenol with $K = 20$ mg/l, $K_{IS} = 1.5$ mg/l, $\hat{q} = 10$ mg/mg VSS$_a$-d, $Y = 0.6$ g VSS$_a$/g, $b = 0.15$/d, and $X_a = 1$ mg VSS/l.

The second type of inhibition is *competitive,* and a separate inhibitor is present at concentration I ($M_T L^{-3}$). The competitive inhibitor binds the catalytic site of the degradative enzyme, thereby excluding substrate binding in proportion to the degree to which the inhibitor is bound. The only parameter affected by $I$ in competitive inhibition is $K_{\text{eff}}$:

$$K_{\text{eff}} = K\left(1 + \frac{I}{K_I}\right) \qquad \textbf{[3.66]}$$

where $K_I$ = an inhibition concentration of the competitive inhibitor ($M_I L^{-3}$). A small value of $K_I$ indicates a strong inhibitor. The rate reduction caused by a competitive inhibitor can be completely offset if $S$ is large enough, because $\hat{q}_{\text{eff}}$ remains equal to $\hat{q}$. Competitive inhibitors usually are substrate analogues.

One example of competitive inhibition that is of interest in environmental engineering practice is that of cometabolism of trichloroethylene (TCE) by microorganisms that grow on substrates such as methane. Here, the first step in methane utilization is its oxidation to methanol by an enzyme called a methane monooxygenase (MMO), which replaces one hydrogen atom attached to the methane carbon with an -OH group. With TCE, MMO interacts to place an oxygen atom between the two carbon atoms to form an epoxide. The key factor is that methane and TCE compete for the same enzyme. The presence of TCE affects the rate at which methane is consumed and, in turn, the presence of methane affects the reaction rate of MMO with TCE. This is illustrated in Figure 3.8. The presence of 20 mg/l TCE reduces the rate of methane utilization considerably over that given by the Monod model, which does not involve competitive inhibition. Similarly, the rate of TCE utilization is greatly reduced as the methane concentration increases. The rate expression for TCE is similar to that for methane, except that the roles of the substrate and the inhibitor are reversed.

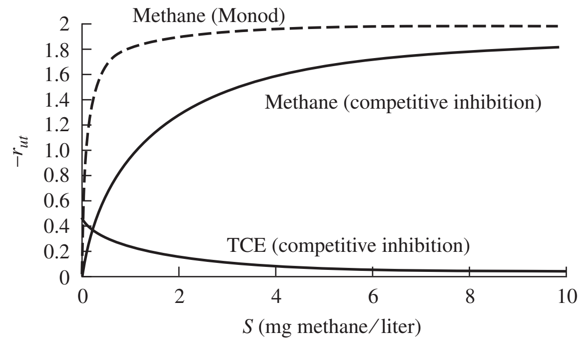


**Figure 3.8** Reaction rates for methane and TCE as governed by competitive inhibition kinetics. Shown for comparison is the rate of methane oxidation in the absence of inhibition. $X_a = 1$ mg $VSS_a$/l, $\hat{q}$ (methane) = 2 mg/mg $VSS_a$-d, $K$ (methane) = 0.1 mg/l, $\hat{q}$ (TCE) = 0.5 mg/mg $VSS_a$-d, $K_I$ (TCE) = 2 mg/l, and $I$ (TCE) = 20 mg/l.

A third type of inhibition is *noncompetitive inhibition* by a separate inhibitor. A noncompetitive inhibitor binds with the degradative enzyme (or perhaps with a coenzyme) at a site different from the reaction site, altering the enzyme conformation in such a manner that substrate utilization is slowed. The only parameter affected is $\hat{q}_{\text{eff}}$:

$$\hat{q}_{\text{eff}} = \frac{\hat{q}}{1 + I/K_I} \qquad \textbf{[3.67]}$$

In the presence of a noncompetitive inhibitor, high $S$ cannot overcome the inhibitory effects, since the maximum utilization rate is lowered for all $S$. This phenomenon is sometimes called allosteric inhibition, and allosteric inhibitors need not have any structural similarity to the substrate.

Figure 3.9 illustrates the different effects that competitive and noncompetitive inhibitors have on reaction rates, in comparison with the Monod model. The values of $I$, $K$, and $K_I$ are assumed to be the same, 1 mg/l, making the value $(1 + I/K_I) = 2$. With a competitive inhibitor, the impact is primarily on $K$, and the inhibitor causes the effective $K$ to increase (from 1 to 2 as indicated by the horizontal line in the middle of the graph). If the substrate concentration ($S$) is high enough, the reaction rate eventually approaches $\hat{q}$. With the noncompetitive inhibitor, the effect is on $\hat{q}$, causing the apparent value to decrease as the inhibitor concentration increases (from 2 to 1 as indicated by the upper and middle horizontal lines in the illustration). The value of $K$, the substrate concentration at which the rate is one-half of the maximum value, in effect, remains unchanged (shown as 1 in the figure by the vertical line). By noting which coefficient in the Monod reaction ($K$ or $\hat{q}$) appears to change when an

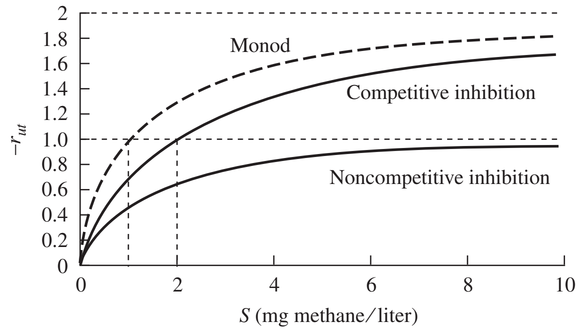


**Figure 3.9** Effect of noncompetitive versus competitive inhibition on reaction kinetics. $K = 1$ mg/l, $K_I = 1$ mg/l, $X_a = 1$ mg VSS$_a$/l, $\hat{q} = 2$ mg/mg VSS$_a$-d, and $I = 1$ mg/l.

inhibitor is added, one can determine whether the inhibitor is acting in a competitive or a noncompetitive manner.

In some cases, competitive and noncompetitive impacts occur together. This situation is termed *uncompetitive inhibition*. Both effective parameters, $\hat{q}_{\text{eff}}$ and $K_{\text{eff}}$, vary as they do for the individual cases:

$$\hat{q}_{\text{eff}} = \frac{\hat{q}}{1 + I/K_I} \tag{3.68}$$

$$K_{\text{eff}} = K(1 + I/K_I) \tag{3.69}$$

*Mixed inhibition* is a more general form of uncompetitive inhibition in which the $K_I$ values in Equations 3.68 and 3.69 can have different values.

The last inhibition type considered is *decoupling*. Decoupling inhibitors, such as aromatic hydrocarbons, often act by making the cytoplasmic membrane permeable for protons. Then, the proton-motive force across the membrane is reduced, and ATP is not synthesized in parallel with respiratory electron transport. Sometimes, decouplers are called protonophores. A decrease in $Y_{\text{eff}}$ and/or an increase in $b_{\text{eff}}$ can model the effects of decoupling inhibition:

$$Y_{\text{eff}} = \frac{Y}{1 + I/K_I} \tag{3.70}$$

$$b_{\text{eff}} = b(1 + I/K_I) \tag{3.71}$$

The other parameters are not necessarily changed. An interesting aspect of decoupling is that the rate of electron flow to the primary acceptor and per unit active biomass can increase. This is shown mathematically for steady state:

$$r_{A,\,\text{eff}} = \frac{\hat{q}_{\text{eff}} S}{K_{\text{eff}} + S}\left[1 - Y_{\text{eff}}\left(1 - \frac{f_d b_{\text{eff}} \theta_x}{1 + b_{\text{eff}} \theta_x}\right)\right] \tag{3.72}$$

in which $r_{A,\,\text{eff}}$ = specific rate of electron flow to the acceptor ($M_s M_x^{-1} T^{-1}$), and all units of mass are proportional to electron equivalents (e.g., COD). Equation 3.72 is valid for all types of inhibition and shows how self-, competitive, and noncompetitive inhibitions normally slow $r_{A,\,\mathrm{eff}}$. However, $r_{A,\,\mathrm{eff}}$ increases when decoupling increases $b_{\mathrm{eff}}$ and/or decreases $Y_{\mathrm{eff}}$.

Sometimes products of a reaction act as inhibitors. The classic example of product inhibition is in alcohol fermentation, where ethanol is produced as an end product from sugar fermentation. In wine manufacture, fermentation of sugar can occur until the ethanol concentration reaches 10 to 13 percent, at which point the alcohol becomes toxic to the yeast, and fermentation stops. This is why the alcohol content of wines normally is in the range of 10 to 13 percent.

## 3.10 Other Alternate Rate Expressions

The Monod model for microbial growth and substrate utilization, as represented by Equations 3.1 and 3.6, respectively, is the most widely used model for kinetic analysis and reactor design. However, other models sometimes are used for special circumstances. The previous section reviewed models for inhibition. This section reviews alternate models when inhibition is not the issue.

One popular alternative is the Contois model, which is represented by

$$r_{\mathrm{ut}} = -\frac{\hat{q}S}{BX_a + S}X_a \qquad [3.73]$$

in which $B$ = a constant [$\mathrm{M_s M_x^{-1}}$]. The Contois equation shows a dependence of the specific reaction rate on the concentration of active organisms present. A high organism concentration slows the reaction rate, which approaches a first-order reaction that depends on $S$, but not $X_a$:

$$-r_{\mathrm{ut}} = \frac{\hat{q}}{B}S \qquad X_a \to \infty \qquad [3.74]$$

The Contois equation is useful for describing the rate of hydrolysis of suspended particulate organic matter, such as are present in primary and waste activated sludges (Henze et al., 1995). It has been noted that hydrolysis rates of biodegradable sludge particles tend to follow first-order kinetics, as is shown by comparing Equations 3.57 and 3.74. The hydrolysis rate is relatively independent of microorganism concentration, even at fairly small organism concentrations. This independence may occur because extracellular enzymes, not the bacteria, carry out the hydrolysis reactions. For hydrolysis, typical values for the ratio $\hat{q}/B$ are on the order of 1 to 3 $\mathrm{d^{-1}}$.

Two other alternate equations are the Moser and Tessier equations, shown by Equations 3.75 and 3.76, respectively.

$$r_{\mathrm{ut}} = -\frac{\hat{q}S}{K + S^{-\gamma}}X_a \qquad [3.75]$$

$$r_{\mathrm{ut}} = -\hat{q}\left(1 - e^{S/K}\right)X_a \qquad [3.76]$$

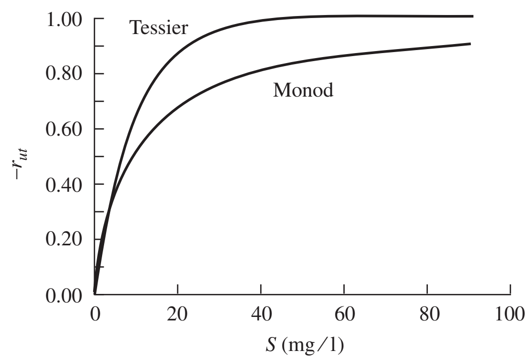


**Figure 3.10** Relationship between the reaction rate and $S$ for the Tessier and Monod equations.

in which $\gamma$ is a constant [unitless]. Like the Contois model, the Moser model reverts to the Monod relationship when $-\gamma = 1$ (or $BX_a = K$ for Contois). The Tessier equation, however, provides quite a different response, as illustrated in Figure 3.10. When the same $K$ and $\hat{q}$ are used in both equations, they show similar responses for $S$ near zero, but the Tessier equation approaches the maximum rate much more quickly than does the Monod equation.

One last alternative rate form is used when more than one substrate is rate limiting, a situation called *dual limitation*. The most common way to represent dual limitation is by the *multiplicative Monod* approach, shown in Equation 3.77:

$$r_{\text{ut}} = -\hat{q}\frac{S}{K+S}\frac{A}{K_A+A}X_a \tag{3.77}$$

in which $A$ = the concentration of the second substrate [$M_A L^{-3}$], and $K_A$ = the half-maximum-rate concentration for the second substrate [$M_A L^{-3}$]. In most cases, the second substrate is the electron acceptor, which is the reason for the symbol $A$. Bae and Rittmann (1996) demonstrated through fundamental biochemical studies that the multiplicative Monod model accurately represents dual limitation by the donor and acceptor. A key feature of Equation 3.77 is that the rate slows in a Monod fashion if either $S$ or $A$ is below saturation, or zero-order. When both substrates are below saturation, the rate decreases by the product of the Monod fractions and can become quite small. For example, if $S = 0.1K$ and $A = 0.1K_A$, $r_{\text{ut}}$ is only 1 percent of $\hat{q}X_a$.

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## 3.12 PROBLEMS

**3.1.** A wastewater is treated in an aerobic CSTR with no recycle. Determine $\theta_x$ min, $[\theta_x^{\min}]_{\lim}$, and $S_{\min}$ if the following parameters hold: $K = 50$ mg/l, $\hat{q} = 5$ mg/mg $VSS_a$-d, $b = 0.06$/d, $Y = 0.60$ g $VSS_a$/g, $S^0 = 220$ mg/l, and $f_d = 1.0$ or 0.8 (two answers).

**3.2.** Consider a chemostat operating at a detention time of 2 h. The following growth constants apply: $\hat{q} = 48$ g/g $VSS_a$-d; $Y = 0.5$ g$VSS_a$/g; $K = 100$ mg/l; $b = 0.1$/d; and $f_d = 1.0$

(*a*) What are the steady-state concentrations of substrate ($S$) for influent substrate concentrations ($S^0$) of 10,000; 1,000; and 100 mg/l?
(*b*) What are the steady-state concentrations of cell ($X_a$) for each $S^0$?
(*c*) What are the steady-state soluble-microbial product concentrations for each $S^0$?
(*d*) What is the minimum detention time below which washout would occur for each $S^0$? What is the limiting value of this washout detention time?

**3.3.** You have been treating a wastewater with bacteria that have the following kinetic coefficients:

$$\hat{q} = 16 \text{ mg BOD}_L/\text{mg VSS}_a\text{-d}$$
$$K = 10 \text{ mg/l}$$
$$b = 0.2/\text{d}$$
$$Y = 0.35 \text{ mg VSS}_a/\text{mg}$$
$$f_d = 0.8$$

A genetic engineer comes to you and claims he can "fix" your bacteria so they are more efficient. The fixed bacteria will have $b = 0.05/\text{d}$ and $f_d = 0.95$, while all other coefficients will be the same.

Using a chemostat reactor, you were successfully using a safety factor of 30, and you plan to continue doing so. How would the genetic engineer's plan affect your sludge production ($QX_v$) and oxygen requirements ($\Delta\, O_2/\Delta_t$)? (Assume $X^0 = 0$ and $S^0 = 300$ mg $\text{BOD}_L$/l). In what ways are the new bacteria more efficient?

**3.4.** Tabulate the value of $S$, $X_a$, $X_i$, SMP, and $S +$ SMP for a chemostat when $\theta_x = 1$, 5, 10, and 30 d. Use the following data:

$$\hat{q} = 16 \text{ mg BOD}_L/\text{mg VSS}_a\text{-d}$$
$$K = 20 \text{ mg/l}$$
$$b = 0.15/\text{d}$$
$$Y = 0.5 \text{ mg VSS}_a/\text{mg}$$
$$f_d = 0.8$$
$$X_i^0 = 50 \text{ mg VSS}_i/\text{l}$$
$$S^0 = 400 \text{ mg BOD}_L/\text{l}$$

and the SMP parameters of Example 3.1.

**3.5.** An industry has a two-stage treatment for its wastewater. The first stage is a chemostat with $\theta = 1$ d. The second stage is a lagoon, which can be approximated as a chemostat with a detention time of 10 d. The kinetic parameters are

$$Y = 0.7 \text{ g VSS}_a/\text{g}$$
$$\hat{q} = 18 \text{ g/g VSS-d}$$
$$K = 10 \text{ mg BOD}_L/\text{l}$$
$$b = 0.25/\text{d}$$
$$f_d = 0.9$$

Assume that the SMP formation can be ignored completely. The total flow rate is 3,785 $m^3$/d, and $S^0 = 1{,}000$ mg $BOD_L$/l.

If $X_v^0 = 40$ mg/l, $X_a^0 = 0$, and all input COD and VSS are biodegradable, estimate the effluent quality (COD, $BOD_L$, VSS) from the whole system. Compute the required $N$ and $P$ supplies; express the answer in mg/l of wastewater flow.

**3.6.** The kinetics of substrate utilization sometimes are approximated by the Eckenfelder relation, $-r_{ut} = k' X_a S$. Solve for the values of $S$ and $X_a$ in a chemostat reactor using the Eckenfelder relation.

**3.7.** An industry has an aerated lagoon for treatment of its wastewater. The wastewater has the characteristics:

$$Q = 10^4 \text{ m}^3/\text{d}$$
$$S^0 = 200 \text{ mg BOD}_L/\text{l}$$
$$X_a^0 = 0$$
$$X_i^0 = 30 \text{ mg VSS}_i/\text{l}$$

The lagoon is well aerated and has a total volume of $4 \times 10^4$ $m^3$. Tracer studies have shown that the lagoon can be described as two CSTRs in series. Therefore, assume that the lagoon is divided into two CSTRs in series. Compute $S$, $X_a$, and $X_i$ in the effluent from both CSTRs.
Use the following parameters:

$$\hat{q} = 5 \text{ mg BOD}_L/\text{mg VSS}_a\text{-d}$$
$$K = 350 \text{ mg BOD}_L/\text{l}$$
$$b = 0.2/\text{d}$$
$$Y = 0.40 \text{ mg VSS}_a/\text{mg BOD}_L$$
$$f_d = 0.8$$

**3.8.** Compute the effluent COD and $BOD_L$ for a chemostat having

$$\theta = 1 \text{ d}$$
$$S^0 = 1{,}000 \text{ mg/l BOD}_L.$$
$$X_a^0 = X_i^0 = 0$$
$$\hat{q} = 10 \text{ mg BOD}_L/\text{mg VSS}_a\text{-d}$$
$$K = 10 \text{ mg BOD}_L/\text{l}$$
$$b = 0.1/\text{d}$$
$$Y = 0.5 \text{ mg VSS}_a/\text{g BOD}_L$$
$$f_d = 0.8$$

What is the percent removal of COD? $BOD_L$? Do not forget all types of effluent COD and $BOD_L$.

**3.9.** You have developed a novel biological treatment process that removes BOD from wastewater by using heterotrophic bacteria that utilize $SO_4^{2-}$ as the electron acceptor. The process is novel and exciting because the $SO_4^{2-}$ is reduced to elemental sulfur ($S(s)$), which ought to be capturable as a valuable end product. Note from the acceptor half-reaction

$$SO_4^{2-} + 6e^- + 8H^+ \rightarrow S(s) + 4H_2O$$

that formation of each mole of $S(s)$ consumes 6 $e^-$ eq, or each gram of $S(s)$ produced consumes 1.5 g of oxygen demand.

You know the following kinetic and stoichiometric parameters for the bacteria carrying out the reaction:

$f_s^0 = 0.08e^-$ eq to cells/$e^-$ eq from donor

$Y = 0.057$ g $VSS_a$/g $BOD_L$

$\hat{q} = 8.8$ g $BOD_L$/g $VSS_a$-d

$b = 0.05$/d

$K = 10$ mg $BOD_L$/l

$f_d = 0.8$

$k_1 = 0.18$ g $BOD_p$/g $BOD_L$

$k_2 = 0.2$ g $BOD_p$/g $VSS_a$-d

$\hat{q}_{UAP} = 1.8$ g $BOD_p$/g $VSS_a$-d

$\hat{q}_{BAP} = 0.1$ g $BOD_p$/g $VSS_a$-d

$K_{UAP} = 100$ mg $BOD_p$/l

$K_{BAP} = 85$ mg $BOD_p$/l

Note that $BOD_p$ represents the $BOD_L$ of the soluble microbial products. Your test wastewater has

$Q = 1{,}000$ m$^3$/d

$S^0 = 500$ mg $BOD_L$/l

$X_i^0 = X_a^0 = 0$

To assess the feasibility of your process, you must compute:

(*a*) If the safety factor is 10, what is the chemostat volume ($V$) in m$^3$?
(*b*) What is the rate of total VSS production in kg/d?
(*c*) What is the effluent $BOD_L$ of organic materials? [Note that effluent S($s$) could exert a BOD, but it is not organic.]
(*d*) What is the rate of S($s$) production in kg S/d?

**3.10.** A wastewater is found to have the following characteristics:

| | | |
|---|---|---|
| COD: | soluble | 100 mg/l |
| | particulate | 35 mg/l |
| $BOD_L$: | soluble | 55 mg/l |
| | particulate | 20 mg/l |
| Suspended Solids: | volatile | 20 mg/l |
| | nonvolatile | 10 mg/l |

Estimate $S^0$ in mg $BOD_L$/l and $X_i^0$ in mg SS/l.

**3.11.** The rate of consumption of a rate-limiting substrate was found to be 4 mg/d per mg of cells when $S = 5$ mg/l and 9 mg/d per mg cells when $S = 20$ mg/l. What are $\hat{q}$ and $K$?

**3.12.** (*a*) Determine the effluent concentrations of $S$, $X_a$, $X_i$, $X_v$, and $X$ (the total suspended solids) for a 10 $m^3$ reactor operated as a chemostat that is fed 3 $m^3$/d of a wastewater containing $BOD_L$ of 2,000 mg/l, 20 mg/l of nonbiodegradable suspended organic solids, and 15 mg/l of inorganic suspended solids. Assume aerobic conditions exist and that $Y = 0.65$ mg cells/mg $BOD_L$, $\hat{q} = 16$ mg $BOD_L$/d per mg cells, $K = 25$ mg/l, $b = 0.2$/d, and $f_d = 0.8$.

(*b*) Calculate the percent substrate ($S$) removal efficiency by the reactor.

**3.13.** You are interested in estimating the maximum growth rate for bacteria ($\hat{\mu}$) for aerobic oxidation of acetate. You have found that the yield of bacteria is 0.45 g per gram of acetate. Also, you have determined that when the acetate concentration is 5 mg/l the rate of utilization is 3 g acetate/g bacteria-d and when the concentration is 15 mg/l, the rate of utilization is 5 g acetate/g bacteria-d. Estimate $\hat{\mu}$.

**3.14.** Assume you are treating a wastewater and that $\hat{q} = 10$ g $BOD_L$/g $VSS_a$-d, $K = 10$ mg $BOD_L$/l, and $b = 0.08\ d^{-1}$. Say that when operating at $\theta_x$ of 4 d the efficiency of substrate removal is 99 percent and oxygen consumption by the microorganisms is found to be 5,000 kg/d, which is determined to be equivalent to an $f_e$ of 0.55. Estimate the oxygen consumption if $\theta_x$ is doubled to 8 d.

**3.15.** A wastewater being considered for biological treatment has the following average values obtained for your evaluation:

| | |
|---|---|
| $BOD_5$(total) | = 765 mg/l |
| $BOD_5$(soluble) | = 470 mg/l |
| $k_1$ | = 0.32 $d^{-1}$ |
| COD: | |
| Total | = 1,500 mg/l |
| Suspended Solids | = 620 mg/l |
| Suspended Solids Concentration: | |
| Total | = 640 mg/l |
| Volatile | = 385 mg/l |

From this information, estimate $S^0$ (mg $BOD_L$/l), $X^0$, $X_v^0$, $X_{in}^0$, and $X_i^0$ (mg/l) for the wastewater. Note that $X_{in}^0$ is the inorganic (or nonvolatile) suspended solids.

**3.16.** The characteristics of a wastewater were found to be the following:

| | |
|---|---|
| $Q$ | = 150,000 $m^3$/d |
| $X$ | = 350 mg/l |
| $X_v$ | = 260 mg/l |
| COD (total) | = 880 mg/l |
| COD (soluble) | = 400 mg/l |
| $BOD_L$ (total) | = 620 mg/l |
| $BOD_L$ (soluble) | = 360 mg/l |

Estimate the following: $S^0$, $Q^0$, $X^0$, and $X_i^0$.

**3.17.** A Haldane modification of the Monod reaction is often used to describe the kinetics of substrate utilization when the substrate itself is toxic at high concentrations (phenol is an example of this). Calculate the effluent concentration for a substrate exhibiting Haldane kinetics from a CSTR and compare this with the value if the substrate were not inhibitory ($S/K_I = 0$) for the following conditions:

$$
\begin{aligned}
V &= 25 \text{ m}^3 \\
Q &= 10 \text{ m}^3/\text{d} \\
S^0 &= 500 \text{ mg/l} \\
Y &= 0.5 \text{ mg/mg} \\
\hat{q} &= 8 \text{ g/g-d} \\
K &= 7 \text{ mg/l} \\
b &= 0.2/\text{d} \\
K_I &= 18 \text{ mg/l}
\end{aligned}
$$

**3.18.** You have evaluated the rate of aerobic degradation of organic matter in an industrial wastewater and found that it does not follow normal Monod kinetics. After some testing you have found the substrate degradation rate seems to follow the relationship:

$$-r_{\text{ut}} = \hat{q}^{0.5} X_a S^{0.5}$$

where

$$
\begin{aligned}
-r_{\text{ut}} &= \text{degradation rate in mg/l-d} \\
\hat{q} &= 4 \text{ l/mg-d}^2 \\
S &= \text{substrate concentration in mg/l} \\
X_a &= \text{active microorganism concentration in mg/l}
\end{aligned}
$$

You have also determined coefficients of bacterial growth on this substrate under aerobic conditions to be as follows:

$$
\begin{aligned}
Y &= 0.25 \text{ g cells/g substrate} \\
b &= 0.08/\text{d}
\end{aligned}
$$

Estimate the effluent substrate concentration when treating the above wastewater in a CSTR while operating at a $\theta_x$ of 4 d.

**3.19.** Fill out the following table to indicate what change an increase in each variable listed would have on the given operating characteristics of an aerobic CSTR. Assume all other characteristics of the wastewater and the other listed variables remain the same. Use: (+) = increase, (−) = decrease, (0) = no change, and (±) = not determinant.

| Variable | Operating Characteristic | | |
|---|---|---|---|
| | $\theta_x$ (days) | Sludge Production (kg/d) | Oxygen Consumption (kg/d) |
| $Y$ | | | |
| $Q$ | | | |
| $S^0$ | | | |
| $b$ | | | |
| $X_i^0$ | | | |

**3.20.** A biological treatment plant (CSTR) is being operated with a $\theta_x$ of 8 d and is treating a waste consisting primarily of substrate $A$. The effluent concentration of $A$ is found to be 0.3 mg/l, and its affinity constant ($K$) is known to be 1 mg/l. A relatively small concentration of compound $B$ is then added to the treatment plant influent and $\theta_x$ is maintained constant. Compound $B$ is partially degraded by the microorganisms present through cometabolism, but acts as a competitive inhibitor to substrate $A$ degradation. If the effluent concentration of compound $B$ is then found to be 1.5 mg/l and its affinity constant ($K_I$) is 2 mg/l, what will now be the effluent concentration of Substrate $A$?

**3.21.** An aerobic CSTR operating at $\theta_x$ of 8 d is treating a wastewater consisting primarily of soluble organic compound $A$. Under steady-state operation the effluent concentration is 1.4 mg/l and its $K$ value is 250 mg/l. Compound $B$ is then added to the wastewater and the reactor volume is increased appropriately to maintain the same $\theta_x$. Different microorganisms use each of the two different substrates. Under the new steady-state conditions, the effluent concentration of $B$ is found to equal 0.8 mg/l. What would you then expect the effluent concentration of compound $A$ to be for the following conditions:

(*a*) Normal Monod kinetics apply for each substrate.
(*b*) Competitive inhibition between substrates occurs, and $K_I$ (related to compound $B$ concentration) is 0.8 mg/l.
(*c*) Noncompetitive inhibition between substrates occurs, and $K_I$ (related to compound $B$ concentration) is 0.8 mg/l.
(*d*) Compound $B$ exhibits substrate toxicity (Haldane kinetics), and $K_{IS} =$ 0.8 mg/l.

**3.22.** The initial rate of oxidation of an organic chemical $A$, when added to an adapted microbial mixed culture (1,000 mg volatile suspended solids/l), was as follows as a function of chemical A concentration:

| | | | | | |
|---|---|---|---|---|---|
| Concentration ($S$), mg/l | 1 | 2 | 4 | 8 | 16 |
| Reaction rate ($-r_{ut}$), mg/l-d | 1.5 | 2.0 | 2.0 | 1.5 | 1.0 |

What rate equation might be used to describe the relationship between reaction rate ($-r_{ut}$) and substrate concentration ($S$)? Why?

**3.23.** You wish to design a biological treatment system for iron contained in drainage water from mine tailings. The water contains no particulate material. You wish to consider aerobic oxidation of the soluble Fe(II) contained in the wastewater to Fe(III), which you will then precipitate in a subsequent reactor by adjustment of pH. For Fe(II) oxidation, $[\theta_x^{min}]_{lim} = 2.2$ d, $f_s^0 = 0.072$, and $b = 0.1$/d.

(*a*) Estimate the detention time for a CSTR reactor with an SF of 10, and $X_v$ of 1,000 mg/l.
(*b*) Estimate the efficiency of Fe(II) oxidation if $S^0 = 10$ mg/l and $K = 0.8$ mg/l.
(*c*) Write a stoichiometric equation for the reaction occurring.
(*d*) Estimate the grams oxygen required per gram Fe(II) oxidized. The molecular weight of Fe is 55.8.
(*e*) Estimate the grams of ammonia-N per gram Fe(II) oxidized that should be added to satisfy bacterial needs.

**3.24.** Methanotrophic bacteria are being used to degrade a mixture of trichloroethylene (TCE) and chloroform (CF) by cometabolism. Here, the enzyme MMO initiates the oxidation of these two compounds. Since they compete with one another for MMO, the rate of their degradation in the mixture is governed by competitive inhibition kinetics. Under the following conditions and with $X_a = 120$ mg/l, estimate the individual rates of transformation of TCE and CF in mg/l-d

$$S_{TCE} = 6.4 \text{ mg TCE/l}$$
$$\hat{q}_{TCE} = 0.84 \text{ gTCE/gVSS}_a\text{-d}$$
$$K_{TCE} = 1.5 \text{ mg TCE/l}$$
$$S_{CF} = 5 \text{ mg CF/l}$$
$$\hat{q}_{CF} = 0.34 \text{ gCF/gVSS}_a\text{-d}$$
$$K_{CF} = 1.3 \text{ mg CF/l}$$

Assume $K_I$ for each substrate equals its respective $K$ value.